

REVIEW

Open Access



Funder's characteristics: a systematic literature review on crowdfunding and the application of the TCCM framework to the equity context

Marco Barone^{1*} , Candida Bussoli¹ and Lucrezia Fattobene²

*Correspondence:
m.barone@lum.it

¹ Department of Management,
Finance and Technology, LUM
Giuseppe Degennaro University,
Casamassima, Italy

² Department of Management
and Law, University of Rome 'Tor
Vergata', Rome, Italy

Abstract

The fintech revolution has empowered consumers to participate in crowdfunding, fostering the collaborative creation of innovative products and services. Despite this progress, the success rate of crowdfunding projects remains low, highlighting the need to understand the factors influencing investors' decisions. Previous systematic literature reviews (SLRs) have revealed a significant gap in the study of funders' characteristics, particularly in equity-based crowdfunding (EC). By conducting a specific SLR guided by the Scientific Procedures and Rationales for Systematic Literature Reviews (SPAR-4-SLR) protocol, this study aims to investigate whether the identified gap is due to a difference in focus or an objective shortfall. The review identified 42 empirical articles, primarily from high-impact economic and managerial journals, with a notable emphasis on technological and sustainability campaigns. The descriptive analysis of the sample highlights the significant presence of articles focused on the characteristics of EC investors. By applying the theories-context-characteristics-methods (TCCM) framework to the EC subset, our analysis reveals a substantial concentration of psychology-related theories, especially concerning demographic factors such as gender and age, alongside psychological determinants such as trust and self-efficacy. By identifying these key characteristics and their theoretical underpinnings, this research advances the understanding of funders' behavior in ECs. The findings suggest that future research should adopt a multidisciplinary approach that incorporates experimental methodologies, such as consumer neuroscience, to capture the cognitive processes influencing investment decisions. This approach can provide deeper insights, enabling crowdfunding platforms and project creators to optimize their strategies and improve campaign success rates.

Keywords: Crowdfunding, Equity-based crowdfunding, Funders' characteristics, Systematic literature review, SPAR-4-SLR, TCCM framework

Introduction

The ongoing fintech revolution is granting consumers a new role in economic and financial systems, allowing them to participate in the collaborative manufacturing process by becoming funders through crowdfunding. Crowdfunding involves a large group of people (the crowd) providing financial resources (funding) to projects proposed by creators, who can be individuals or organizations (Mollick 2014; Popescul et al. 2020). This approach allows startups, which often struggle to access traditional funding channels such as banks or financial markets (Robb and Robinson 2014), to secure financial support and bridge the early-stage funding gap (Hemer et al. 2011). It also transforms the role of consumers, giving them access to innovative products and direct influence on product development, thereby enabling the creation of new offerings (Belleflamme et al. 2014; Chaney 2019). Owing to crowdfunding, consumers can discover and support cutting-edge technologies, unique designs, and creative projects early in their development. This involvement allows consumers to actively participate in the innovation process (Stanko and Henard 2016).

Despite the rapid growth of the crowdfunding market, the success rate of crowdfunding projects remains relatively low (Zhu et al. 2023). Four systematic literature reviews (SLRs) have examined success factors in crowdfunding, each with distinct objectives. Hoegen et al. (2018) focus on factors influencing investment decisions in crowdfunding, comparing them to traditional investments such as venture capital and bank loans. Mazzocchi and Lucarelli (2023) explore the literature on EC to understand its determinants of success. Neuhaus et al. (2022) specifically investigate the personality traits of entrepreneurs, whereas Deng et al. (2022) identify determinants affecting crowdfunding campaign success and the impact of various success measurements and crowdfunding models.

While all these SLRs address crowdfunding, none specifically focus on funders' characteristics. From their extensive work, it becomes clear that there is a scarcity of research on funder characteristics, with even fewer studies exploring these characteristics in relation to ECs. Despite the growing importance of crowdfunding in corporate financing and the still existing need for a better understanding of crowd investors' behavior (Wang and Prokop 2024), few empirical studies have addressed this topic. Thus, our SLR is directly aimed at assessing studies that address funders' characteristics within the crowdfunding context, shedding light on their role in the process. Understanding individual traits, including factors such as proximity, is crucial for successfully raising funds through any crowdfunding platform, as these traits significantly influence investor attention and decision-making, ultimately impacting crowdfunding outcomes. Cognitive psychology theories suggest that cognitive architectures shape how individuals respond to stimuli (Anderson 1982), emphasizing the link between cognitive skills and perceived risk in financial decisions (Burks et al. 2009; Rustichini et al. 2016). Additionally, sociodemographic variables—such as gender, age, and income—play a vital role in investment decisions (Barber and Odean 2001; Kumar and Goyal 2016). Research indicates that these factors affect both the types of projects to which investors are drawn and their willingness to contribute financially (Manocha et al. 2023; Shao et al. 2004). Therefore, the primary aim of this study is to conduct a systematic literature review specifically focusing on crowdfunding funders' characteristics.

Investors receive rewards for their contributions in various ways, resulting in different types of crowdfunding. In reward-based crowdfunding, project creators offer future in-kind rewards in exchange for backer contributions, whereas in ECs, when a new business is established, the creator permits consumers to purchase shares, allowing them to become investors and earn dividends. There is a tendency in the literature to develop analyses in the context of reward-based crowdfunding, linked to a greater and earlier diffusion of the two most popular platforms, Kickstarter and Indiegogo (Stiver et al. 2015; Zhao et al. 2019). Recently, EC has gained prominence, with the global transaction value reaching \$65 billion in 2022 and an expected annual growth rate of 1.9% from 2017 to 2027 (Statista 2023). It has also become particularly attractive to academics, as it closely resembles traditional long-term investments (Popescul et al. 2020). Prior research indicates the importance of understanding the characteristics of crowdfunding investors in the EC context (Josefy et al. 2017; McKenny et al. 2017). Consequently, numerous authors highlight the necessity of grasping the characteristics and motivations that distinguish fund providers (Bruton et al. 2015; Collins and Pierrakis 2012; Lukkarinen et al. 2019). Despite this recognition, the literature often overlooks the specific characteristics of funders themselves. To address this oversight, our study explores several critical aspects. We investigate the extent to which ECs are underrepresented in analyses that consider funder characteristics; examine the evolution of the literature concerning EC investors in terms of theories, research context, characteristics, and methods; and contemplate future directions for research in this area. By examining these dimensions, we seek to offer valuable insights that enhance the understanding of the role funders play in the crowdfunding landscape.

This study contributes to the literature addressing the absence of research on decision makers' individual characteristics by systematically reviewing the literature on the influence of funder characteristics on campaign success. We seek to shed light on this critical variable and analyze whether there is insufficient focus on this topic within the context of EC, a segment that is essential for financial innovation. By employing the theories-context-characteristics-methods (TCCM) framework to review the literature, we provide actionable insights into existing gaps, particularly in underexplored areas. This approach not only clarifies the state of research on funder characteristics in ECs but also offers targeted recommendations for future studies, ensuring that our contribution significantly advances the field.

Specifically, there is a call to construct a comprehensive set of multidisciplinary theories that address the attributes of fundraisers, funders, campaign pages, and other involved third parties. This suggestion aims to incorporate an analysis that encompasses signal sets from external sources (bottom-up approach) and consumers' cognitive structures (top-down). At this juncture, it is advised to embrace a more meaningful adoption of experimental methodology, leveraging innovations and rigor offered by neuroscience, for instance. A consumer neuroscience approach can effectively capture unconscious and automatic processes that traditional techniques may not adequately address (Plassmann et al. 2012).

The following SLR is disclosed in seven more paragraphs. Second, the theoretical framework of the literature through the definition of crowdfunding is outlined, and previous SLRs on the topic are analyzed. In the third paragraph, the topic of EC and

the importance of identifying the characteristics of funders are developed, and on these considerations, the research questions are constructed. The fourth section describes the systematic literature search methodology adopted, indicating the search strategy. The fifth section outlines the bibliometric and thematic results inherent to the entire sample under investigation (all the crowdfunding categories). The sixth and seventh paragraphs present the results of using the TCCM framework on the subsample related to the EC and provide suggestions for future research. Finally, the conclusions constitute the eighth paragraph.

Background

Previous research

This section summarizes the four main SLRs on the determinants of crowdfunding success.

The SLR by Deng et al. (2022) aims to systematically investigate the factors influencing crowdfunding campaign success, identifying 94 empirical papers that examine its determinants. The authors underscore the increasing importance of crowdfunding in the academic literature, as demonstrated by the increase in publications from 2011 to 2021. Additionally, most of these papers focus on reward-based crowdfunding and are published in the fields of economics and finance. They highlight the lack of a unified analytical approach to understanding the determinants of crowdfunding project success. The dependent variables used to measure success vary widely, encompassing eight different constructs, both dichotomous (e.g., funding success) and continuous (e.g., funds raised, success ratio, number of investors). This diversity also extends to the methodologies employed. Despite this variance, researchers have predominantly utilized linear regressions, logit/probit/tobit regressions, and PLS-SEM (for questionnaires). The use of a diverse set of independent or determinant variables leads to considerable variability in the analyses. Deng et al. (2022) identify four categories of factors influencing the success of a campaign: project-related, campaign creator, platform, and funder factors. The project-related category is most affected by the proliferation of independent variables, encompassing characteristics and soft information associated with the project. Among the 94 papers analyzed, 78 utilized the minimum campaign goal, 56 used the campaign duration, 56 employed the campaign category, and 29 used other determinants. The factors inherent to the campaign creator (e.g., experience, geography, gender, education, and others) and those related to the platform (e.g., type, age, competition) also constitute a significant focus of studies. However, features related to backers or investors (e.g., motivations, experience, geography, and affective reactions) are present in only six out of the 94 papers considered in the analysis. The authors argue that factors related to funders or investors merit further analysis, as do non-reward-based projects. Only one paper in the examined literature focuses on the characteristics of investors in EC campaigns.

Hoegen et al. (2018) focused their review on understanding and visualizing decision-making processes and their influencing factors in crowdfunding, analyzing 68 articles primarily concentrated in the final years of their study period, which extends to 2016. There are many commonalities with Deng et al. (2022), including the paucity of articles related to EC (seven) and investor characteristics (10): the conjunction of equity-based

crowdfunding and investor characteristics categories produced only six articles. Hoegen et al. (2018) assume that funders' characteristics should be considered moderators in combination with the other categories of determinants, described as context, financial and campaign statistics, project and product quality, creator attributes, and social communities. However, the authors recognize the importance in the crowdfunding investment decision of investors' cognitive characteristics, such as their need for cognition or cognitive heuristics, citing work such as Gigerenzer and Goldstein (1996) and Tversky and Kahneman (1974). Again, a greater focus on investor characteristics is recommended for future research.

Mazzocchi and Lucarelli (2023) aimed to provide a multidisciplinary framework for understanding the reasons behind the success or failure of equity-based crowdfunding campaigns, drawing insights from the literature produced between 2015 and 2022. They identify 32 articles in which only two focus on investor characteristics: Li et al. (2016), through the presence or absence of lead investors, and Vulkan et al. (2016), which identifies positive and negative relationships between different investor profiles and the probability of campaign success. On the other hand, the goal of the two authors is to provide a framework to guide future research in identifying the signals conveyed by entrepreneurs and team creators, placing the information provided in the categories of hard (firm characteristics, financial information, business characteristics and project description) and soft (intellectual capital, human capital, social capital and social media network) information. Mitigating information asymmetries also entails uncovering the external factors identified in campaigns and platform characteristics necessary to create trust between entrepreneurs and investors. This direction is also evident in suggestions for future research. For the authors, research should consider investor characteristics as a dependent variable and look for those factors that determine the investor's propensity to invest in EC campaigns, as well as their attitudes and preferences toward perceived risk.

Neuhaus et al. (2022) highlight how expressed personality traits influence investors' decisions in crowdfunding, highlighting the growing scholarly interest in analyzing success factors in campaigns. As with all nascent literature, the authors acknowledge a bias among researchers toward specific characteristics of crowdfunding campaigns. Indeed, Neuhaus et al. (2022) divide the literature into streams that identify evidence regarding crowdfunding campaigns' characteristics and those concerning entrepreneurs' demographic features. The authors report consistent findings across all the papers: studies examining entrepreneur personality consistently demonstrate a significant relationship with success in crowdfunding (e.g., the Big Five, self-efficacy, innovativeness, locus of control, and need for achievement). However, even this SLR found only one article (Stevenson et al. 2019b, a) that included investor characteristics when analyzing the success factors of EC campaigns.

Overall, previous SLRs clearly highlight the underdevelopment of research exploring funders' characteristics, particularly in the context of ECs.

Significance of funders' characteristics

Previous SLRs seem to denote an absence of a research stream that considers funders' characteristics in crowdfunding campaigns. Nevertheless, individual conditions unequivocally influence the process of investor attention and decision-making, likely

affecting crowdfunding success. Cognitive psychology theories of skill acquisition suggest that cognitive architectures modify the response of different individuals to similar stimuli (Anderson 1982). A crucial role of cognitive processes has also been observed for financial consumer decisions, e.g., the association between cognitive skills and assumed risk (Burks et al. 2009; Rustichini et al. 2016). In addition to psychological variables, sociodemographic variables, e.g., gender, age, and income (Barber and Odean 2001; Kumar and Goyal 2016), are crucial in investment decisions (Manocha et al. 2023; Shao et al. 2004).

The absence of research on decision-makers' individual characteristics, despite their potential relevance for behavior, prompts a critical question: Is this lack merely apparent, stemming from the diverse aims of previous SLRs, or does it indeed signify a research gap on the topic? Given the lack of SLRs specifically addressing the role of funders' characteristics in campaign success, the primary aim of this study is to shed light on this topic through the following research question:

RQ1 What is the current state of the literature regarding the influence of funders' characteristics on campaign success?

Moreover, given the relevance of EC and the findings of previous systematic literature reviews on research underproduction on this topic and following the protocol titled 'Scientific Procedures and Rationales for Systematic Literature Reviews (SPAR-4-SLR)' (Paul et al. 2021), we intend to answer the following questions:

RQ2 Is EC underrepresented in crowdfunding analyses that consider funder characteristics?

RQ3 How does the literature on EC investors evolve with respect to theories, research context, characteristics, and methods?

RQ4 Where should EC research be headed?

Research methodology

Structure of review

In line with the purpose of this study, an SLR is conducted. The SLR is a rigorous method for performing structural investigations of a field of study (Munn et al. 2018), which is believed to be reliable and significant (Higgins et al. 2019), and for identifying gaps in a research domain, allowing researchers to provide future research directions (Gopalakrishnan and Ganeshkumar 2013; Paul and Criado 2020). Among the various protocols proposed in the literature, the new SPAR-4-SLR protocol by Paul et al. (2021) is followed. The TCCM organizing framework by Paul and Rosado-Serrano (2019) is integrated to answer RQ3 and RQ4.

Search strategy, data assembly and purification

In accordance with SLRs published in top-ranked journals (Roy Bhattacharjee et al. 2022) or within similar domains (Deng et al. 2022; Hoegen et al. 2018; Mazzocchi and

Lucarelli 2023; Mendes et al. 2019; Neuhaus et al. 2022), the Web of Science (WoS) and EBSCOhost (EBSCO) have been employed as online search engines. In addition, the search process was cross-checked with Google Scholar to ensure that all the articles were included.

Figure 1 shows the keywords used to intercept the final sample of journal articles. The selection of keywords was aimed at delimiting the scope of the research as defined in the research questions. The search string consists of three parts connected by the Boolean 'AND'; the 'OR' command intercepts all variants that refer to similar concepts, and asterisks (*) indicate singular and plural words or extensions. To answer RQ1 and RQ2, the keywords used are generally connected with all the crowdfunding typologies. Specifically, the first part of the string refers to the main context of analysis, crowdfunding.

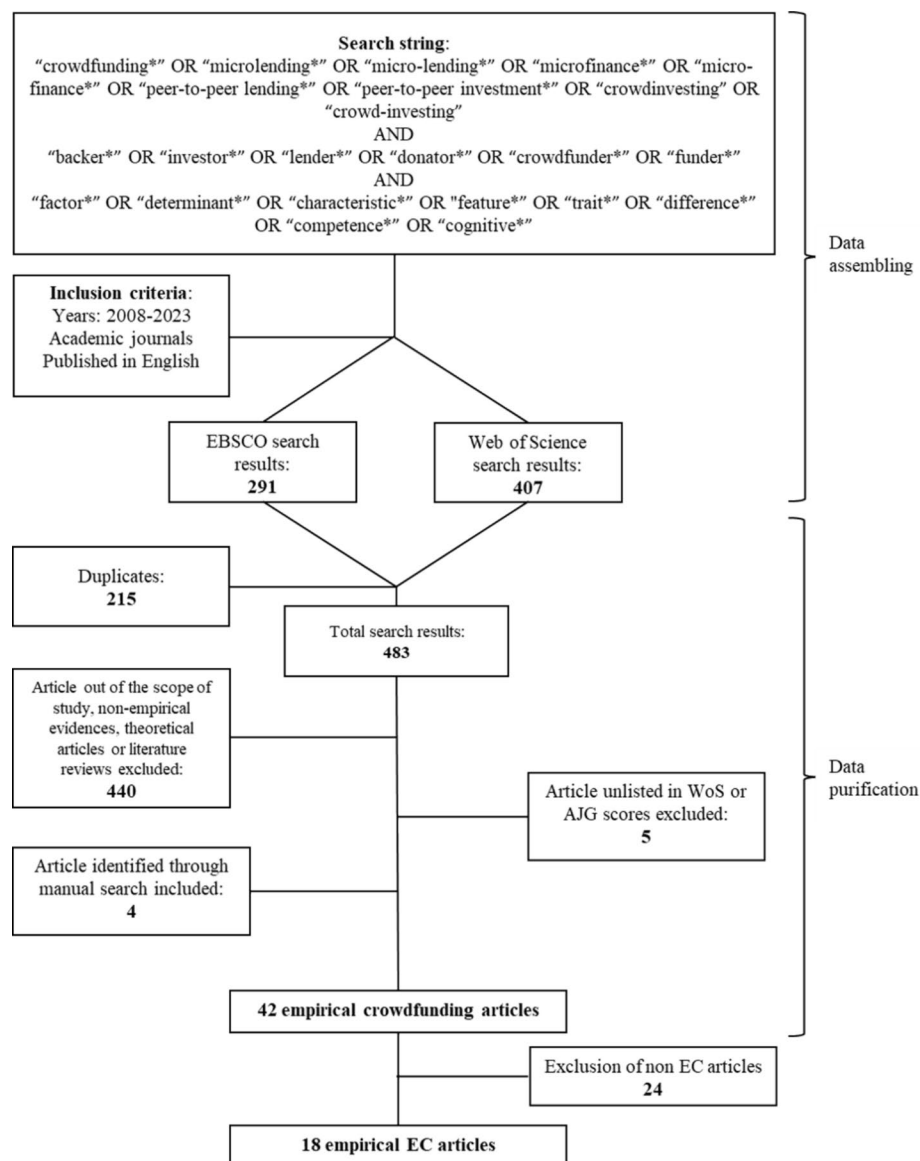


Fig. 1 Article selection process

Following Deng et al. (2022) and Neuhaus et al. (2022), keyword selection uses all the crowdfunding-related words. The second part of the string aims to detect the articles that study the backer's/investor's perspective and includes all words connected with the synonym of the crowdfunding customer. Finally, the last section focuses on the characteristics or determinants related to the funders. It derives from a search performed via the Google Scholar platform (Talwar et al. 2020) using 'investor characteristics' as a search word. After the analysis of the main citation results, we hand-collected the following list: "factor*" OR "determinant*" OR "characteristic*" OR "feature*" OR "trait*" OR "difference*" OR "competence*" OR "cognitive*". Keywords had to be present in the title or the abstract of the articles for the papers to be detected (Deng et al. 2022).

The inclusion criteria set the starting point of the research at 2008, aligning with the advent of two major crowdfunding platforms (Stiver et al. 2015; Zhao et al. 2019), and ended in July 2023. The second inclusion criterion relates to the search for articles published in academic journals, regardless of their Academic Journal Guide (AJG) rating, that have undergone a peer-review process, excluding 'gray literature', books, chapters, conference proceedings, dissertation abstracts and working papers (Danese et al. 2018; Mazzocchi and Lucarelli 2023). Finally, only articles published in the English language were included in the sample of papers for review (Deng et al. 2022; Mazzocchi and Lucarelli 2023). At the end of this first step, 407 articles were obtained through WoS, and 291 were obtained from EBSCO. Matching between the two databases identified and excluded 215 duplicates, resulting in a sample of 483 units.

The analysis was performed according to the exclusion criteria, which involved reviewing both the abstracts and full texts of the articles. This step aimed to exclude articles that did not present an empirical approach (e.g., Ahsan et al. 2018; Hapsari et al. 2022; Maehle 2020; Olufolaji and Phillips 2015) and literature reviews (e.g., Brannon et al. 2022; Hoegen et al. 2018; Mazzocchi and Lucarelli 2023). Moreover, articles that used no variables referring to funders' sociodemographic, psychological or personal characteristics in their empirical analyses were excluded. After the exclusion of 440 papers, an additional five papers were removed (e.g., Hanif and Santosa 2023) because the journals in which they were published are not on either the lists on WoS rankings or in the AJG by the Chartered Association of Business Schools.

Four articles derived from other SLRs or expert knowledge were included (Davis et al. 2017; Giudici et al. 2020; Greenberg and Mollick 2017, ; Stevenson et al. 2019b, a). The total sample consists of 42 empirical articles, which generally analyze a single funder's characteristics (Table 1).

To answer the third and fourth RQs, a subsample of 18 EC articles was extracted.

Results for all crowdfunding categories

To answer RQ1 and RQ2, a two-step process was conducted on the 42 articles (Table 1). The first descriptive analysis aims to delineate trends, geographic distributions, and research focal points within the literature concerning the characterization of backers/investors across different types of crowdfunding. The second part consists of thematic analysis of the articles to identify the knowledge gained about the characteristics of funders that contribute to the success of the crowdfunding campaign. This initial analysis addresses the general attributes of funders across various campaign types, including

Table 1 Characteristics of the articles

N°	Article title	Author(s)	Year	Journal title	Field of Journal	Country of University	Platform Name	Platform Type
1	To buy or to back? Backers' motivations for fashion crowdfunding projects	Ahn, Sookyoung	2023	Journal of Fashion Marketing and Management: an international journal	Business; management	South Korea	Undefined	Reward
2	Investor intention in equity crowdfunding. Does trust matter?	Alharbey, Mohammed; Van Hemmen, Stefan	2021	Journal of Risk and Financial Management	Business, finance	Spain; Saudi Arabia	Manafa	Equity
3	The impact of financial literacy on the investment decision of non-donation-based crowdfunding in Indonesia	Arifah, Jihan Nadra; Dalimunthe, Zuliani	2020	International Journal of Business and Society	Business	Indonesia	Undefined	Generic (nondonation)
4	Gender gaps in equity crowdfunding: evidence from a randomized field experiment	Bapna, Sofia; Ganco, Martin	2021	Management Science	Management; operations research and management science	USA; USA	EquityCrowdfund	Equity
5	Community-level social capital and investment decisions in equity crowdfunding	Barbi, Massimiliano; Febo, Valentina; Giudici, Giancarlo	2023	Small Business Economics	Economics; business; management	Italy; France; Italy	Italian platforms	Equity
6	Crowdfunding and social influence: an empirical investigation	Cecere, Grazia; Le Guel, Fabrice; Rochelandet, Fabrice	2017	Applied Economics	Economics	France; France; France	Undefined	Reward; Donation
7	Financial literacy and gender difference in loan performance	Chen, Jia; Jiang, Jiajun; Liu, Yu-jane	2018	Journal of Empirical Finance	Business, finance; economics	China, China	Renrendai	Lending
8	Tackling gender bias in equity crowdfunding: an exploratory study of investment behavior of Latin American investors	Cicchello, Antonella Francesca; Kazemikhastagh, Amirreza	2022	European Business Review	Business	Italy; Mexico	Broota.com.br; Eqsed; Start Me Up; EuSocio; Broota.cl; Crowdfunder.mx; Play Business	Equity

Table 1 (continued)

N°	Article title	Author(s)	Year	Journal title	Field of Journal	Country of University	Platform Name	Platform Type
9	Household preferences for investing in crowd-funding	Giobotaru, Laurentiu-Cristian; Kim, Sul; van Soest, Arthur	2021	De Economist	Economics	Netherlands	Undefined	Equity
10	Funders' positive affective reactions to entrepreneurs' crowdfunding pitches: the influence of perceived product creativity and entrepreneurial passion	Davis BC, Hmieleski KM, Webb JW, Coombs JE	2017	Journal of Business Venturing	Business	USA; USA; USA	Kickstarter	Reward
11	Toward an understanding of online lending intentions: evidence from a survey in China	Dongyu Chen; Hao Lou; Van Slyke, Craig	2015	Communications of the Association for Information Systems	Computer science; information systems	China; USA; USA	PPDai	Lending
12	Segmenting "digital investors": evidence from the Italian equity crowdfunding market	Feola, Rosangela; Vesci, Massimiliano; Marinato, Ezio; Parente, Roberto	2021	Small Business Economics	Economics; business; management	Italy	Italian platforms	Equity
13	Elective affinities: exploring the matching between entrepreneurs and investors in equity crowdfunding	Giudici, Giancarlo; Guerini, Massimiliano; Rossi-Lamastra, Cristina	2020	Baltic Journal of Management	Management	Italy	Italian platforms	Equity
14	Crowdfunding in entrepreneurial projects: disentangling patterns of investor behavior	Goethner, Maximilian; Luettig, Sebastian; Regner, Tobias	2021	Small Business Economics	Economics; business; management	Germany	Companisto	Equity
15	Activist choice homophily and the crowdfunding of female founders	Greenberg, Jason; Molllick, Ethan	2017	Administrative Science Quarterly	Business and management	USA; USA	Kickstarter	Reward

Table 1 (continued)

N°	Article title	Author(s)	Year	Journal title	Field of Journal	Country of University	Platform Name	Platform Type
16	Women backing women: the role of crowdfunding in empowering female consumer-investors and entrepreneurs	Groza, Mya Pronschinske; Groza, Mark D.; Barral, Luis Miguel	2020	Journal of Business Research	Business	USA, SPAIN	Lanzanos	Reward
17	Discrimination in lending markets: status and the intersections of gender and race	Harkness, Sarah K	2016	Social Psychology Quarterly	Psychology	USA	Undefined	Lending
18	Determinants of individual investment decisions in investment-based crowdfunding	Herve, Fabrice; Manthe, Elodie; Sannajust, Aurelie; Schwienbacher, Armin	2019	Journal of Business Finance and Accounting	Business, Finance	France; France; France	WISEED	Equity; Lending
19	Attracting prosocial lenders from different cultures to help others in microlending	Hong, JungHwa (Jenny); Byun, Kyung-Ah (Kay)	2020	Journal of Consumer Marketing	Business	USA	Undefined	Lending
20	The local bias in equity crowdfunding: behavioral anomaly or rational preference?	Hornuf, Lars; Schmitt, Matthias; Stenzhorn, Eliza	2022	Journal of Economics and Management Strategy	Economics; Management	Germany; Germany; Germany	Companisto; Innovestment	Equity
21	A woman's place is in the... startup! Crowdfunder judgments, implicit bias, and the stereotype content model	Johnson, Michael A.; Stevenson, Regan M.; Letwin, Chaim R	2018	Journal of Business Venturing	Business	USA; USA; USA	Wefunder.com	Equity
22	Understanding the determinants of funders' investment intentions on crowdfunding platforms a trust-based perspective	Kang, Minghui; Gao, Yiwu; Wang, Tao; Zheng, Haichao	2016	Industrial Management and Data Systems	Computer Science, Interdisciplinary Applications	China; China	Angelcrunch; Zhongchou	Equity

Table 1 (continued)

N°	Article title	Author(s)	Year	Journal title	Field of Journal	Country of University	Platform Name	Platform Type
23	Why funders invest in crowdfunding projects: role of trust from the dual-process perspective	Liang, Ting-Peng; Wu, Shelly Ping-Ju; Huang, Chih-chi	2019	Information and Management	Information Science and Library Science; Management	Taiwan, China; Australia	FlyingV	Reward
24	Untangling the influence of perceived sustainability orientation on value-co-creation behavior in crowdfunding process: investigating a mediation model	Liang, Xiaobei; Hu, Xiaojuan Y.; Li, Eldon; Meng, Hu	2023	Internet Research	Business	China; Taiwan, China; China	Jingdong	Reward
25	What's in a logo? The impact of complex visual cues in equity crowdfunding	Mahmood, Ammara; Luffarelli, Jonathan; Mukesh, Mudra	2019	Journal of Business Venturing	Business	Canada; France; UK	Undefined	Equity
26	Gender differences in the contribution patterns of equity-crowdfunding investors	Mohammadi, Ali; Shafi, Kourosh	2018	Small Business Economics	Economics; Business; Management	Sweden; USA	FundedByMe	Equity
27	In crowdfunding we trust: a trust-building model in lending crowd-funding	Moysidou, Krystallia; Hausberg, J. Plet	2020	Journal of Small Business Management	Management	UK; Germany	Lendix	Lending
28	Fintech attaining sustainable development: an investor perspective of crowdfunding platforms in a developing country	Rizwan, Amina; Mustafa, Faisal	2022	Sustainability	Environmental Studies; Green and Sustainable Science and Technology; Environmental Sciences	Pakistan	Pakistani platforms	Equity
29	Reward crowdfunding contribution as planned behavior: an extended framework	Shneor, Rotem; Munim, Ziaul Haque	2019	Journal of Business Research	Business	Norway, Norway	Mesenaatti	Reward

Table 1 (continued)

N°	Article title	Author(s)	Year	Journal title	Field of Journal	Country of University	Platform Name	Platform Type
30	Out of control or right on the money? Funder self-efficacy and crowd bias in equity crowdfunding	Stevenson, Regan M.; Ciuchta, Michael P.; Letwin, Chaim; Dinger, Jenni M.; Vancouver, Jeffrey B	2019	Journal of Business Venturing	Business	USA; USA; USA; USA	Undefined	Equity
31	Warm-glow giving, hedonism, and their influence on muslim user engagement on loan-based crowdfunding platforms	Sutanto, Juliana; Weninger, Helena; Duriana, Handle	2021	Journal of the Association for Information Systems	Information Science and Library Science; Computer Science, Information Systems	UK; Indonesia	Undefined	Lending
32	Crowdfunding sustainable entrepreneurship: what are the characteristics of crowdfunding investors?	Tenner, Isabell; Horisch, Jacob	2021	Journal of Cleaner Production	Green and Sustainable Science and Technology; Engineering, Environmental Sciences	Germany	Ecocrowd; Leih-DeinerUmweltGeld	Reward; Equity
33	Are you part of the crowd? The role of sex and environmental characteristics for crowdfunding awareness	Vaznyte, Egle; Andries, Petra; Manigart, Sophie	2023	Journal of Small Business Management	Management	Netherlands; Belgium; Belgium	Undefined	Generic (no lending)
34	Reconnoitering antecedents of donation intention in donation crowdfunding campaigns: a mediating role of crowdfunding readiness	Yadav, Ajit; Mathur, Himendu Prakash	2023	International Review on Public and Nonprofit Marketing	Business	India	Milaap; Ketto	Donation
35	Linguistic style and gender match in funding intention toward crowdfunding project	Wan Mohamad Nazarie, Wan Nur Fazni; Williams, Russell	2021	Review of International Business and Strategy	Business	Malaysia; UK	Generosity-GoFundMe	Donation

Table 1 (continued)

N°	Article title	Author(s)	Year	Journal title	Field of Journal	Country of University	Platform Name	Platform Type
36	Understanding backers' funding intention in reward crowdfunding: an elaboration likelihood perspective	Wang, Zhengpei; Yang, Xue	2019	Technology in Society	Social Issues; Social Sciences, Interdisciplinary	China; China	Sojump	Reward
37	Influence of perceived risks on the decision to invest in equity crowdfunding: a study of Malaysian investors	Wasiuzzaman, Shaista; Chong, Lee Lee; Ong, Hway Boon	2022	Journal of Entrepreneurship in Emerging Economies	Business	Brunei Darussalam; Malaysia	Malasian platforms	Equity
38	Developing and validating a theory-based model of crowdfunding investment intention: perspectives from social exchange theory and customer value perspective	Yang, Xiaocong; Zhao, Kai; Tao, Xiaobo; Shiu, Eric	2019	Sustainability	Environmental Studies; Green and Sustainable Science and Technology; Green and Sustainable Science and Technology; Environmental Sciences	China; China; China; UK	Undefined	Undefined
39	Crowdfunding preferences for a sustainable milk product with integrated photovoltaic water pumping system in China	Zhang, Chi; Campana, Pietro Elia; Liu, Chengxi; Wang, Ke; Yan, Jinyue	2019	Applied Energy	Energy and Fuel	Sweden; Sweden; Sweden; China	Undefined	Reward
40	Investor participation in reward-based crowdfunding: impacts of entrepreneur efforts, platform characteristics, and perceived value	Zhang, Fengjiao; Zhang, Hong; Gupta, Sumeet	2023	Information Technology and Management	Information Science and Library Science; Management	China; China; India	Jingdong	Reward
41	Backer motivation in crowdfunding new product ideas: is it about you or is it about me?	Zhang, Haisu; Chen, Weizhi	2019	Journal of Product Innovation Management	Business; Management; Engineering, Industrial	USA	Kickstarter	Reward

Table 1 (continued)

N°	Article title	Author(s)	Year	Journal title	Field of Journal	Country of University	Platform Name	Platform Type
42	Determinants of backers' funding intention in crowdfunding: social exchange theory and regulatory focus	Zhao, Qun; Chen, Chun-Der; Wang, Jin-Long; Chen, Pin-Chung	2017	Telematics and Informatics	Information Science and Library Science	China; Taiwan, China; Taiwan, China	FlyingV	Reward

Repetition of nationalities in the University Nationality column refers to different universities in the same country

the primary theories employed, the geographical areas of participants, and the variables used to measure crowdfunding success, ensuring that context is adequately accounted for.

Overview of selected articles

The temporal distribution of articles (Fig. 2) reveals an increasing trend in the publication of topics related to the analysis of funders' characteristics, with a peak recorded in 2019 (nine publications) and six articles already published in the first half of 2023. The lowest value of one was observed in 2015, coinciding with the publication of Chen et al. (2015), which marked the inception of articles related to the systematic review. The lack of articles published between 2008 and 2014 suggests that crowdfunding can be considered a novel phenomenon in consumer research.

As might be expected, the number of papers detected here year-for-year is lower than that reported by Deng et al. (2022) given the narrower research scope of the present study; this also validates our search strategy outcomes. On the other hand, the 42 articles identified confirm the growing importance of crowdfunding in global scientific research, even specifically with respect to funders' characteristics. Hence, the phenomenon under analysis increases the ability to attract entrepreneurs and funders (or donors) to an alternative funding avenue to the traditional financial sector.

The global importance of the topic is also revealed by the high number of nationalities of the universities to which the authors of the 42 articles belong (Fig. 3a). Specifically, 23 different nations are noted for a total of 88 universities.

The USA is the most represented in the sample, with 21 universities, followed by China (16), France (eight), Germany (six), the UK (five), and Italy and Taiwan China (four). However, the total number of European universities (27) exceeds that of the USA. At the bottom of the sample, there are seven countries each with only one university: Australia, South Korea, Mexico, Indonesia, Pakistan, Canada, and Brunei.

Panel b of Fig. 3 shows the countries where the populations under investigation are sampled. In particular, China outperforms the United States of America with nine papers versus seven, and Germany outperforms France with four papers versus two. Italy figures with three papers, ahead of Malaysia, India and Indonesia with one. Excluding the two papers that do not explicitly report this information (Bapna and Ganco 2021; Mahmood

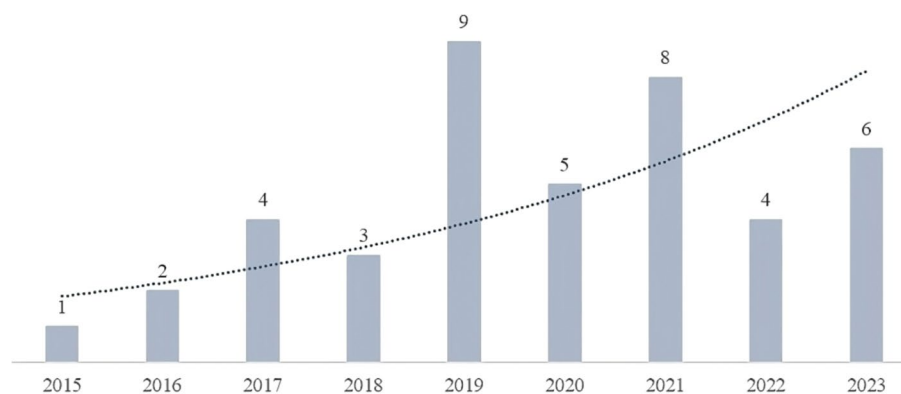


Fig. 2 Article frequency analysis by year

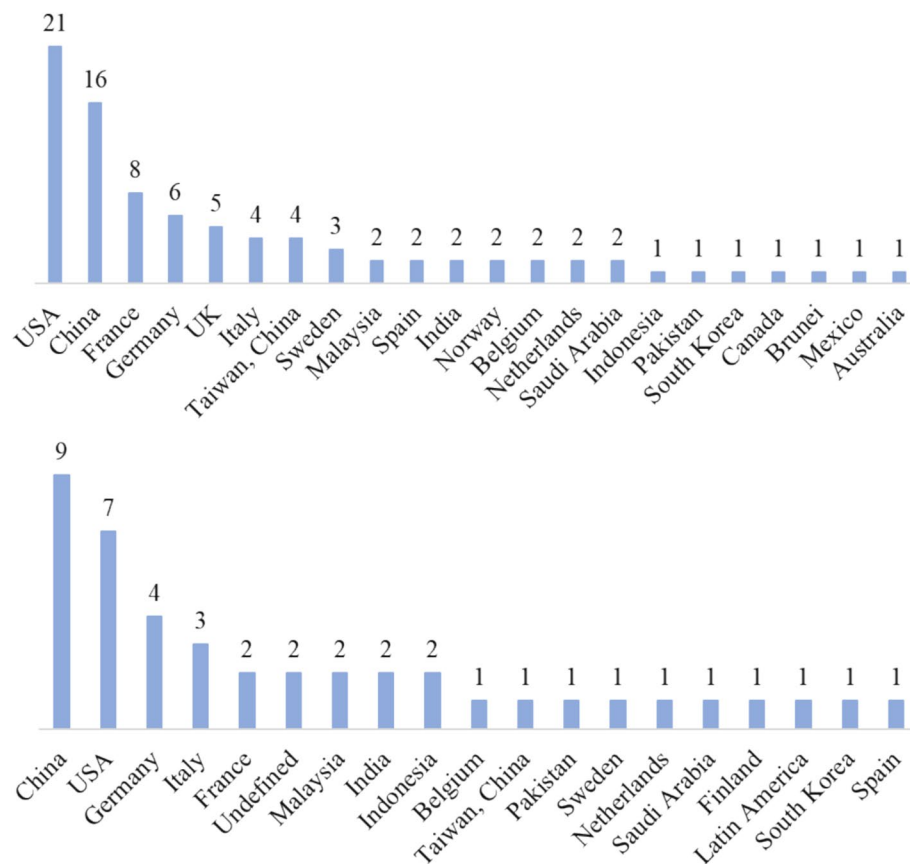


Fig. 3 Breakdown of articles per university (top) and study participants' geographical location (bottom)

et al. 2019), the sample comprises 41 studies spanning 18 countries. The extra unit refers to the work of Hong and Byun (2020), which analyses both the U.S. and Indian contexts.

Authors and journal features

Figure 4 shows the sample composition on the basis of the number of authors per single article, with values ranging from one author to five. There were two studies with one author (i.e., Ahn 2023; Harkness 2016), as well as five (i.e., Stevenson et al. 2019b, a; Zhang et al. 2019). Most papers were produced by three (17 articles; 40.47%) or two authors (14 articles; one-third of the sample). Finally, seven (16.66%) articles had four authors.

Table 2 presents the outlet journals where the 42 articles in the sample were published, along with their WoS impact factors and AJG scores as provided by the Chartered Association of Business Schools. Additionally, the table displays the field categories assigned to the journals by WoS, except for *De Economist*, which was assigned by AJG.

The *Journal of Business Venturing* (Davis et al. 2017; Johnson et al. 2018; Mahmood et al. 2019; Stevenson et al. 2019b, a) and the *Small Business Economics* journal (Barbi et al. 2023; Goethner et al. 2021; Mohammadi and Shafi 2018; Feola et al. 2021) each accounted for the highest number of articles in the sample, with four publications each, collectively covering 19.05% of the sample. These journals are categorized within

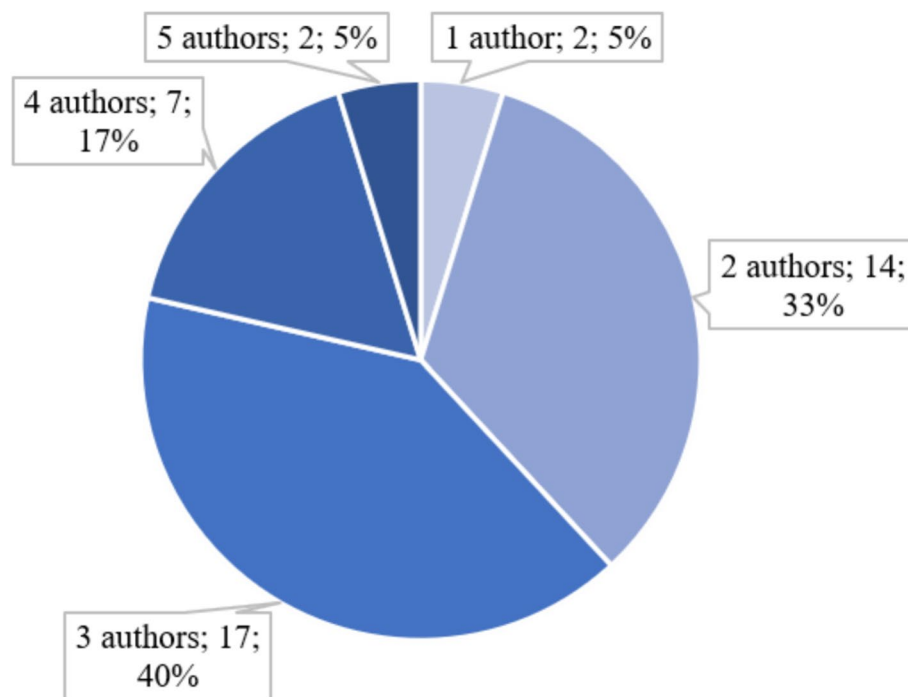


Fig. 4 Number of authors per article

business, economics, and management research areas. Additionally, three other journals are the outlets of more than one publication: the Journal of Business Research (Groza et al. 2020; Shneor and Munim 2019), the Journal of Small Business Management (Moyssidou and Hausberg 2020; Vaznyte et al. 2023), and Sustainability (Rizwan and Mustafa 2022; Yang et al. 2019).

AJG evaluations integrate statistical citation information with peer reviews, editorial, and expert judgments in management and business, resulting in scores ranging from one to four, marked with an asterisk (*). Some journals in the sample are marked as “n.a.” (not available) owing to their absence from AJG’s list. This deficiency is compensated for by including WoS scores. According to AJG’s 2021 ABS, leading journals include Administrative Science Quarterly (Greenberg and Mollick 2017), the Journal of the Association for Information Systems (Sutanto et al. 2021), and Management Science (Bapna and Ganco 2021). The Journal of Business Venturing and the Journal of Product Innovation Management are rated fourth. Eight journals (19.05% of the sample) received the lowest score of one.

WoS impact factors are based on the citation frequency over a specific period. The five journals with the highest values are the Journal of Business Research (11.3), Applied Energy (11.2; Zhang et al. 2019), the Journal of Cleaner Production (11.1; Tenner and Hörisch, 2021), the Journal of Product Innovation Management (10.5), and Administrative Science Quarterly (10.4). Only one journal in the sample has a score slightly below one: the Journal of Risk and Financial Management (Alharbey and Van Hemmen 2021), with a value of 0.95.

The areas related to economic sciences are the most numerous, specifically business (12), management (nine) and economics (five). The number of papers related to the

Table 2 Journal characteristics

Journal Title	N° of articles	ABS2021	Impact Factor WoS	WoS category
Journal of Business Venturing	4	4	8.7	Business
Small Business Economics	4	3	6.4	Economics Business Management
Journal of Business Research	2	3	11.3	Business
Journal of Small Business Management	2	3	6.2	Management
Sustainability	2	n.a	3.9	Environmental studies Green and sustainable science and technology Environmental sciences
Administrative Science Quarterly	1	4*	10.4	Business and Management
Applied Economics	1	2	2.2	Economics
Applied Energy	1	n.a	11.2	Energy and Fuel
Baltic Journal Of Management	1	1	3.1	Management
Communications of the Association for Information Systems	1	2	2	Computer science, information systems
De Economist	1	1	-	Economics**
European Business Review	1	2	4.2	Business
Industrial Management and Data Systems	1	2	5.5	Computer science, interdisciplinary applications
Information and Management	1	3	9.9	Information science and library science Management
Information Technology and Management	1	1	2.4	Information science and library science Management
International Journal of Business and Society	1	n.a	1	Business
International Review on Public and Non-Profit Marketing	1	1	1.7	Business
Internet Research	1	3	5.9	Business
Journal of Business Finance and Accounting	1	3	2.9	Business, finance
Journal Of Cleaner Production	1	2	11.1	Green and sustainable science and technology Engineering, environmental Environmental sciences
Journal of Consumer Marketing	1	1	2.8	Business
Journal of Economics and Management Strategy	1	2	1.9	Economics Management
Journal of Empirical Finance	1	3	2.6	Business, finance Economics
Journal of Entrepreneurship in Emerging Economies	1	1	3.6	Business
Journal of Fashion Marketing and Management	1	1	3.5	Business Management
Journal of Product Innovation Management	1	4	10.5	Business Management Engineering, industrial
Journal of Risk and Financial Management	1	n.a	0.95	Business, Finance

Table 2 (continued)

Journal Title	N° of articles	ABS2021	Impact Factor WoS	WoS category
Journal of the Association for Information Systems	1	4*	5.8	Information science and library science Computer science, information systems
Management Science	1	4*	5.4	Management Operations research and management science
Review of International Business and Strategy	1	1	3.1	Business
Social Psychology Quarterly	1	n.a	2.7	Psychology
Technology in Society	1	n.a	9.2	Social issues Social sciences, interdisciplinary
Telematics and Informatics	1	n.a	8.5	Information science and library science

** Category from AJG

sustainability subject area is also substantial, with contributions from the journal *Sustainability* (two papers) and the *Journal of Cleaner Production* (one paper). Additionally, there are articles on computer science and information systems (Zhao et al. 2017) and psychology (Harkness 2016). Notably, a journal may be attributed to more than one category by WoS, either related (e.g., ‘Economics’, ‘Business’, and ‘Management for Small Business Economics’) or complementary (e.g., ‘Information Science and Library Science’ and ‘Management’ for the journals *Information and Management* and *Information Technology and Management*).

Platforms and projects

The answer to RQ2 can be gleaned from Fig. 5, which highlights the distribution in absolute values of the different types of crowdfunding under analysis in the 42 articles in the sample. The categorization is conservative: ‘Generic (nonlending)’ (Vaznyte et al. 2023), ‘Generic (nondonation)’ (Arifah and Dalimunthe 2020), or undefined (Yang et al. 2019) are not counted as ECs.

The least frequent classes in the sample are donation-based crowdfunding, comprising three articles out of 39 (7.69%), and lending-based crowdfunding, which includes seven articles (approximately 17.95%), which is consistent with Deng et al. (2022). A reversal of positions is detectable for the other two crowdfunding categories. While Deng et al. (2022) identified a significant number of articles about reward-based crowdfunding, the present SLR consists of a majority of research related to equity-based crowdfunding, albeit by a small margin. The number of articles on reward-based crowdfunding that also consider funders’ characteristics in their analysis is 14 out of 39 (35.90%), whereas the number of those on equity-based crowdfunding is 18 (42.86%).

A due explication is to be made regarding the presence of some works (among the 42 in the sample) that operate analyses on two different types of platforms: Cecere et al. (2017) use both reward and donation-based crowdfunding; Tenner and Horisch (2021), both reward and equity-based; and Hervé et al. (2019), equity and lending. However, the

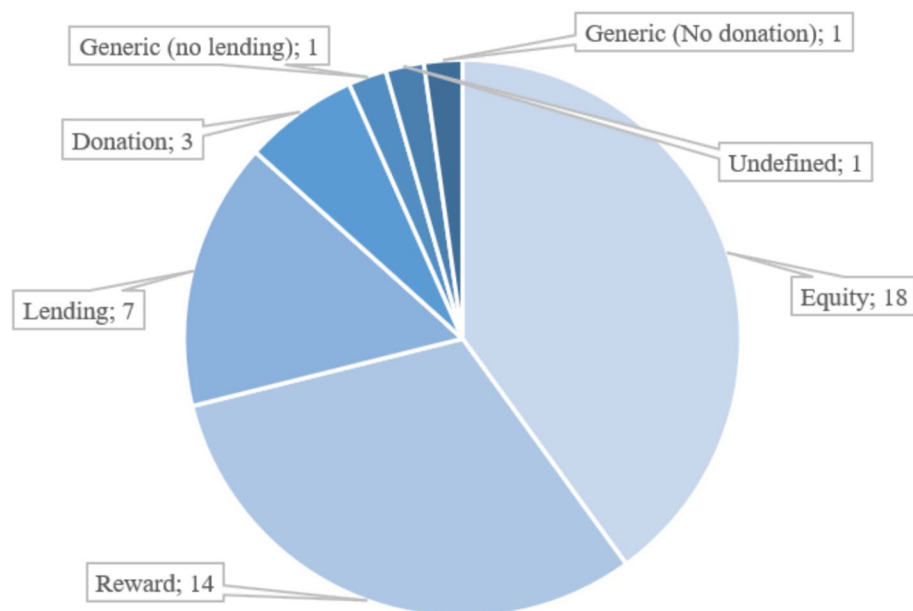


Fig. 5 Crowdfunding type count

distribution of these three articles allows for no imbalance in any of the four categories found in the literature.

Overall, in the literature examining funders' characteristics, participation in entrepreneurial activity through ECs is well represented compared with other types of campaigns. Consistent with the literature, EC has the greatest affinity with traditional investments (Popescu et al. 2020). Understanding this form of crowdfunding requires analyzing investors' decision-making processes, considering personal characteristics such as cognitive factors (Hoegen et al. 2018).

One of these aspects is the campaign theme (Fig. 6). Most articles (20) do not specify the type of campaign being analyzed, especially when coinciding with analyses on the general propensity for funding on crowdfunding projects (e.g., Alharbey and Van Hemmen 2021; Arifah and Dalimunthe 2020; Harkness 2016; Kang et al. 2016; Liang et al. 2019; Yadav and Mathur 2023; Zhao et al. 2017) or generic knowledge of crowdfunding platforms (Vaznyte et al. 2023).

Among the categories explicitly defined in the articles, the most frequent category is 'technology' (15), followed by 'sustainability' (seven), 'fashion and arts' (five), 'entertainment' (five), 'other services' (four), 'consumer products' (two), and 'food and drinks' (two). The sum of the categories exceeds the number of papers in the sample because some papers analyze multiple categories (Hervé et al. 2019; Hornuf et al. 2022; Goethner et al. 2021; Greenberg and Mollick 2017; Mahmood et al. 2019; Yang et al. 2019; Zhang and Chen 2019).

Empirical investigations

An investigation of the methodologies used in the sample (Fig. 7) reveals that the survey is the most commonly used technique for identifying funders' characteristics, with 23 articles accounting for 55% of the sample. Eight articles (approximately 19% of the

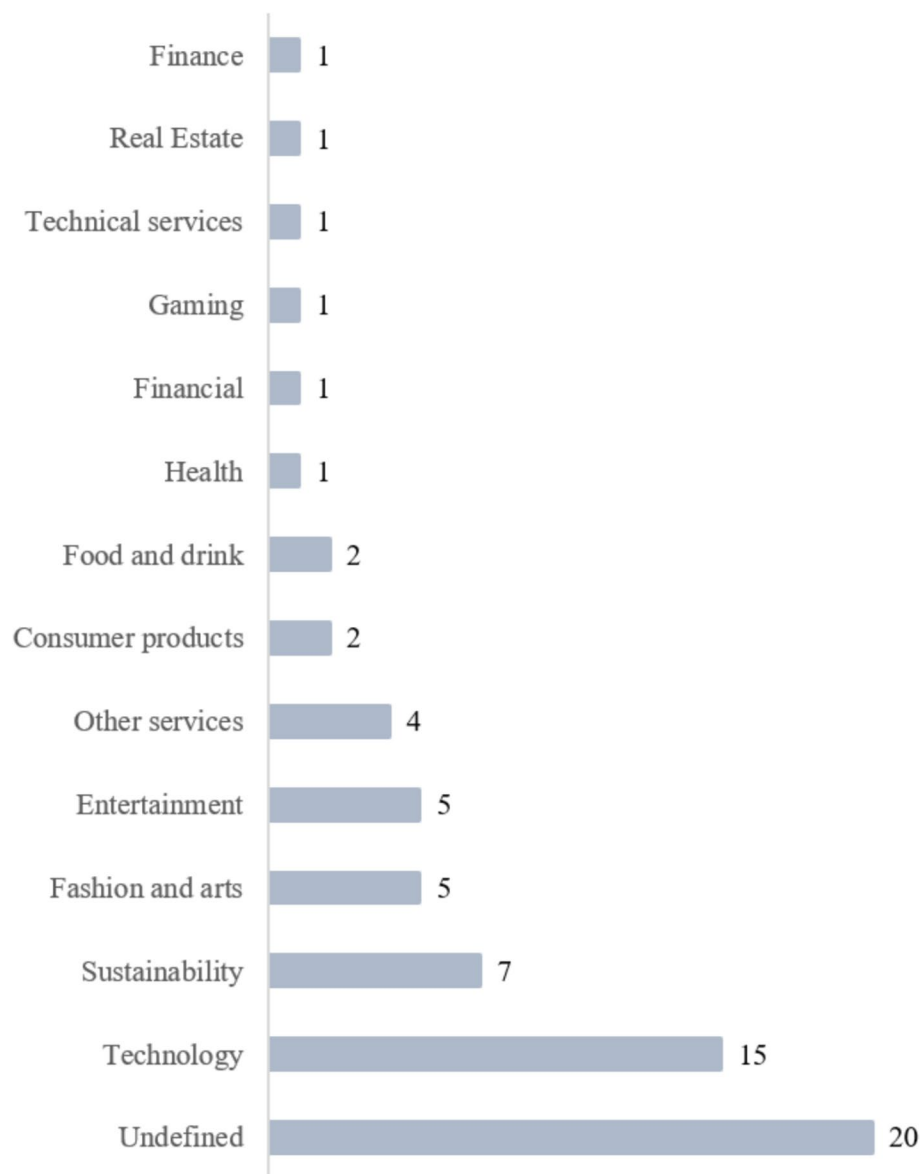


Fig. 6 Crowdfunding campaign themes count

sample) adopted an experimental design. Finally, some empirical analyses use data from preconstructed datasets (e.g., Chen et al. 2018; Hervé et al. 2019; Mahmood et al. 2019; Mohammadi and Shafi 2018) or handmade datasets (e.g., Barbi et al. 2023; Cicchiello and Kazemikhasragh 2022; Giudici et al. 2020; Goethner et al. 2021) directly from information retrieved from the internet.

Two papers combine survey and experimental design, developing multiple studies in the same article (Greenberg and Mollick 2017; Zhang and Chen 2019).

Table 3 summarizes the theories used in the sample of articles under review. Excluding eight articles without theoretical references, the remaining empirical papers use a set of 36 different theories, including Green and Mollick's (2016) own theory (i.e., the Theory of Choice Homophily). Notably, some articles employ multiple theories simultaneously.

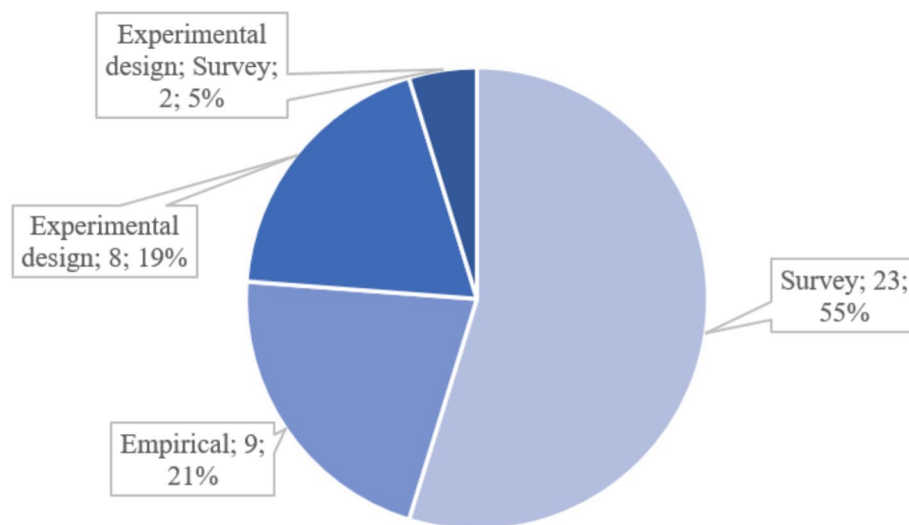


Fig. 7 Methodology used in sample articles

Table 3 Theories adopted in sample articles

N°	Theory	Sum of articles	N°	Theory	Sum of articles
1	None	8	20	Information Gap Theory	1
2	Theory of Planned Behavior	4	21	Information Processing Theory	1
3	Homophily Theory	3	22	Means-End Theory	1
4	Affective Events Theory	2	23	Processing Fluency Theory	1
5	Self-Determination Theory	2	24	Rational Choice Theory	1
6	Social Exchange Theory	2	25	Reward Expectations Theory	1
7	Swift Trust Theory	2	26	Risk Perception Theory	1
8	Theory of Choice Homophily	2	27	Sequential Reciprocity Theory	1
9	Trust Theory	2	28	Social Capital Theory	1
10	Commitment-Trust Theory	1	29	Social Network Theory	1
11	Signaling Theory	1	30	Status Characteristics Theory	1
12	Agency Theory	1	31	Stereotype Content Model	1
13	Perceptual Control Theory	1	32	Theory of Basic Human Values	1
14	Customer Value Perspective	1	33	Theory of Law and Finance	1
15	Discrete Choice Modeling	1	34	Theory of Reasoned Action	1
16	Dual Process Theory	1	35	Theory of Trust Transfer	1
17	Elaboration Likelihood Theory	1	36	Transfer Trust Theory	1
18	Gender Role Theory	1	37	Warm-glow Giving Theory	1
19	Information Asymmetry Theory	1			

For example, Liang et al. (2023) use five different theories (i.e., theory of planned behavior, agency theory, rational choice theory, affective events theory, and risk perception theory); Groza et al. (2020) incorporate social capital theory and the theory of choice homophily; Chen et al. (2015) exploit information asymmetry theory and trust theory; Ahn (2023) uses self-determination theory and information gap theory; and others.

The sample's most frequent (four) theory is the theory of planned behavior (Liang et al. 2023; Rizwan and Mustafa 2022; Shneor and Munim 2019; Yadav and Mathur 2023).

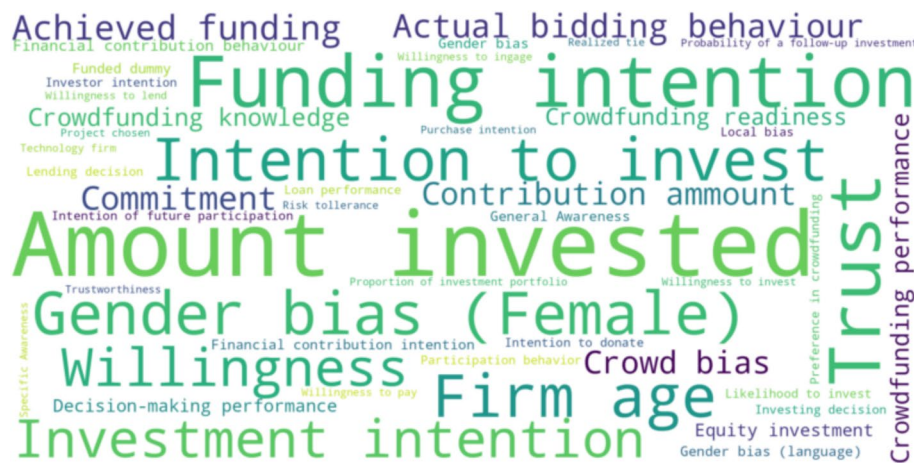


Fig. 8 Dependent variables used

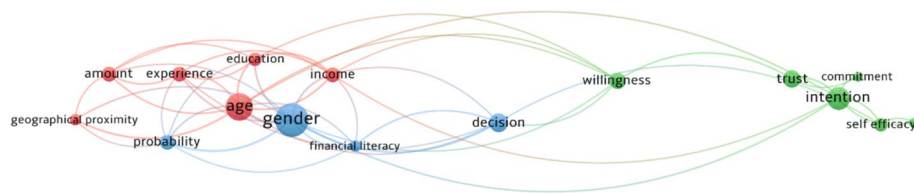


Fig. 9 Visualization of variable networks

The second most common theory in the sample (three) is Homophily Theory (Cicchiello and Kazemikhasragh 2022; Giudici et al. 2020; Wan Mohamad Nazarie and Williams 2021).

Significant variability can be detected in the dependent variables related to the success of crowdfunding campaigns or funders' decisions. Figure 8 shows all the variables used in the sample of selected articles, where word size indicates frequency. The three most frequently used dependent variables refer to the intention to invest/fund/donate/lend (12), the amount invested (six), and the willingness to fund (six) by survey participants or experiments.

Figure 9 presents a visualization of the network of variables considering both dependent and independent variables. The most frequently used predictive variables in the sample of articles are related to sociodemographic information: gender (20), age (13), income (six), education (five) and geographical proximity (four). The other characteristics of funders investigated in the literature are related to psychological or behavioral aspects, such as self-efficacy (five) and trust (three), investment experience (three) or financial literacy (two).

Summary and critical review of funders' characteristic literature in crowdfunding

The literature on funders' characteristics in crowdfunding has seen significant growth over the past eight years, underscoring its increasing relevance in academic discourse and that researchers have started to recognize the relevance of investigating investors'

individual differences and their role in the crowdfunding process. Most publications appear in high-impact, internationally recognized journals, although the research remains heavily concentrated in the fields of economics, business, and management. While the theme of sustainability is gaining traction, there is a notable lack of studies from psychological or neuroscientific perspectives, despite the rising importance of consumer neuroscience (Alvino et al. 2020). This lack of studies is a significant problem because it limits our understanding of the cognitive and emotional factors that drive investor behavior. Without these insights, research may miss crucial elements that influence decision-making, such as cognitive biases, social influences, and emotional responses. This gap can lead to a superficial analysis of funding motivations, resulting in campaigns that fail to connect effectively with potential supporters. As consumer neuroscience becomes increasingly relevant, integrating these insights is essential for creating more effective and resonant approaches in crowdfunding and beyond.

Most analyses tend to rely heavily on surveys and data collection methods rather than experimental designs. This reliance can limit the depth of insights gained, as surveys may not fully capture the complexities of decision-making processes or the nuances of consumer behavior. On the other hand, experimental designs allow for a more controlled examination of causal relationships and can provide stronger evidence for how specific factors influence funding decisions. Overrelying on surveys, the literature may miss opportunities to explore dynamic interactions and underlying motivations in greater detail, ultimately resulting in findings that are less robust and potentially less applicable to real-world scenarios. A balanced approach that incorporates both methodologies could lead to richer, more comprehensive insights into funder behavior.

Research predominantly focuses on equity-based and reward-based crowdfunding, with fewer studies addressing lending and a minimal emphasis on donation-based crowdfunding. Additionally, the literature frequently suffers from imprecise definitions of specific campaign categories, as broad terms such as “technology” can create overlaps between different types of campaigns (e.g., technology in fashion). This lack of clarity can obscure the unique characteristics and challenges associated with each category, making it difficult to draw meaningful comparisons or insights. However, studies that define particular themes often focus on technology and sustainability, both of which hold considerable significance in business and economic contexts. This tendency to prioritize these themes may limit the exploration of other important areas, ultimately constraining the breadth of understanding in crowdfunding research. A more nuanced approach to categorization could enhance the literature by providing clearer frameworks for analyzing diverse crowdfunding initiatives.

The summary of the theories used in the reviewed articles reveals significant diversity, reflecting the complexity of funder behavior in crowdfunding. The Theory of Planned Behavior stands out as the most frequently utilized framework, followed by Homophily Theory, with many studies integrating multiple theoretical approaches.

However, this diversity also raises concerns about theoretical fragmentation. The reliance on various theories within individual studies suggests a lack of consensus on the key factors influencing crowdfunding behavior, which hinders the establishment of a cohesive understanding of the dynamics at play. As a result, researchers may struggle to build upon one another’s findings effectively. A more unified theoretical framework

could improve clarity and foster deeper insights into the mechanisms that drive funder engagement, ultimately advancing the field of crowdfunding research.

The analysis of dependent variables in crowdfunding research shows variability, with the most frequently examined factors being the intention to invest, the amount invested, and the willingness to fund. The key predictive variables are mainly sociodemographic, such as gender, age, income, education, and geographical proximity, alongside some psychological aspects such as self-efficacy and trust. Critically, this focus may limit the understanding of the complex nature of crowdfunding success. While sociodemographic factors are relevant, an overemphasis on them can obscure other important influences, such as motivations and emotional involvement. The limited exploration of psychological factors indicates a missed opportunity to gain deeper insights into funder behavior. A broader approach could enhance models of engagement and campaign success.

The TCCM framework applied to EC studies

This SLR focuses on the category related to EC and applies the TCCM framework (Paul and Rosado-Serrano 2019) to highlight how previous studies have been conducted, with what methodologies, theories applied, contexts considered, and proposals for future research (Paul et al. 2023).

Theories

In the subset of 18 articles focusing on EC, 13 theoretical frameworks can be identified (refer to Table 4). Notably, four articles (Barbi et al. 2023; Ciobotaru et al., 2021; Goethner et al. 2021; Mohammadi and Shafi 2018) do not specify the adoption of any particular theory. In contrast, some authors utilize more than one theoretical framework,

Table 4 Theories applied in research on the characteristics of EC investors

Theory	N° of articles	References	Sample article
None	5	–	Barbi et al. (2023); Ciobotaru et al. (2021); Feola et al. (2021); Goethner et al. (2021); Mohammadi and Shafi (2018)
Homophily Theory	2	Ruef et al. (2003)	Cicchello and Kazemikhasragh (2022); Giudici et al. (2020)
Stereotype Content Model	1	Fiske et al. (2002)	Johnson et al. (2018)
Theory of Basic Human Values	1	Schwartz (1992, 1994)	Tenner and Hörisch (2021)
Social Network Theory	1	–	Hervé et al. (2019)
Theory of Planned Behavior	1	Ajzen (1991)	Rizwan and Mustafa (2022)
Gender Role Theory	1	Heilman (2012)	Bapna and Ganco (2021)
Law and Finance Theory	1	Porta et al. (1998)	Wasiuzzaman et al. (2022)
Swift Trust Theory	1	Meyerson et al. (1996)	Alharbey and Van Hemmen (2021)
Agency Theory	1	Aghion and Bolton (1992)	Hornuf et al. (2022)
Commitment-Trust Theory	1	Morgan and Hunt (1994)	Kang et al. (2016)
Processing Fluency Theory	1	Reber et al. (2004)	Mahmood et al. (2019)
Perceptual Control theory	1	Powers (1973)	Stevenson et al. (2019b, a)
Transfer Trust Theory	1	Stewart (2003)	Alharbey and Van Hemmen (2021)
Total	19*		

* Alharbey and Van Hemmen (2021) use two theories

as observed in Alharbey and Van Hemmen (2021) and Cicchiello and Kazemikhasragh (2022).

The classification of theories according to their research branches involves four areas: psychology, sociology, economics, and social psychology. Importantly, theories may be of interest in more than one branch of research, as they often interact and have implications in several scientific disciplines. The classification provided here attempts to associate theories with the main branches in which they are commonly studied or applied.

Psychological theories

Seven theories can be traced back to psychology and are repeated only once each.

Alharbey and Van Hemmen (2021) explore how EC investors' trust in the platform and the fundraiser are determinants of the intention to invest. The theoretical assumption of the two authors relies on two psychological theories, the *Trust Transfer Theory* and the *Swift Trust Theory*, as the focus is on identifying the mechanism related to the transfer of trust from the institution (in this case, the platform) to the specific project/fundraiser to be funded. Stewart's Trust Transfer Theory (2003) asserts that if two entities are combined, the trust residing in one could also be transferred to the second by a third party (trust is conveyed from the platform to the vendors). The Swift Trust Theory of Meyer-son et al. (1996) predicts the existence of swift trust occurring in temporary organisational structures that have a finite '*life span, formed around a shared and relatively clear goal or purpose, and their success depends on a tight and coordinated coupling of activity*'. According to this theory, a group or team initially assumes trust and then verifies and modifies it appropriately.

The *stereotype content model* (SCM) (Fiske et al. 2002) is used by Johnson et al. (2018) to verify the presence of a gender bias between the entrepreneur's and investor's genders in the EC. Johnson et al. (2018) incorporate trustworthiness and competence perceptions into their analysis to examine whether gender investors may be affected in their investment choice by the perceptions of trust and competence that they attribute to the fundraiser's gender. The SCM proposed by Fiske et al. (2002) asserts that individuals are evolutionarily predisposed to evaluate strangers in terms of their willingness to perform warmth (or help) and competence. All group stereotypes and interpersonal impressions are formed along these two dimensions.

Bapna and Ganco (2021) investigate the gender bias in EC platforms, but they rely on *gender role theory* (Heilman 2012), which suggests stereotyping of occupations and the presence of purely male or female jobs. By applying this theory to EC, Bapna and Ganco (2021) hypothesize an influence on interest in an investment on the basis of the gender of the investor (along with experience) and the fundraiser.

Schwartz's *theory of basic humans* (1992, 1994) provides a comprehensive theory of intercultural psychology and universal values and extends previous schemes of intercultural communication, such as Hofstede's theory of cultural dimensions. Schwartz introduces ten motivational types of values, each distinguished by its underlying motivation or goal, and explains how they are recognizable by individuals from all cultures. Tenner and Horisch (2021) use this theoretical framework to analyze the influence of specific individual values (self-enhancement, self-transcendence, openness-to-change and conservative value) on the probability of investing in sustainable EC projects.

Mahmood et al. (2019) offer a conceptual framework in which the funding decision results from logo complexity, i.e., the subjective level of elaborateness and intricacy of a logo design, but investors' perceptions of venture innovativeness mediate the relationship. For this purpose, they design an analysis on the *basis of processing fluency theory* (Reber et al. 2004), according to which the subjective experience of ease or difficulty with which individuals process a stimulus influences the perceptions, responses, and judgments of individuals to that stimulus.

Stevenson et al. (2019b, a) propose the existence of a relationship between investors' self-efficacy and funding knowledge and decision-making performance in ECs through funder searching efforts. Specifically, self-efficacy is supposed to be negatively related to effort in the assigned task (the investment decision). For this purpose, the authors rely on Powers' (1973) *perceptual control theory*, according to which behavior controls perception: individuals act against perturbations of the environment to align hierarchically organized perceived aspects of the self and the world with neural-specified reference values. Over time, this theory has spread across various disciplines, such as experimental psychology, neuroscience, ethology, anthropology, linguistics, sociology, robotics, developmental psychology, organisational and management psychology and numerous other fields.

Sociological theories

The sociological theories observed in the sample of articles are attributable to *Homophily Theory* (Ruef et al. 2003) and *Social Network Theory*.

Cicchello and Kazemikhasragh (2022) and Giudici et al. (2020) exploit *Homophily theory* to investigate the investment probability of a campaign depending on equalities (or differences) between investors and entrepreneurs in terms of gender (Cicchello and Kazemikhasragh 2022), geographical proximity and age (Barbi et al. 2023).

Hervé et al. (2019) use social network theory—by citing Vismara (2016)—and state that a higher level of social capital affects crowd investor investments. Liu et al. (2017) argue that social network theory focuses on the role of social relationships in spreading information, transmitting personal or media influence, and enabling attitudinal or behavioral change. Hervé et al. (2019) suggest that investors with greater social interactions are more inclined to invest, which is also the case in ECs.

Economic theories

Among the sample of articles on the attributes of EC investors, two economic theories (*law and finance theory and agency theory*) were retrieved in many papers (Hornuf et al. 2022; Wasiuzzaman et al. 2022).

The *Law and Finance Theory* of Rafael La Porta, Florencio Lopez-de-Silanes, and Andrei Shleifer was presented in their academic article entitled 'Law and Finance' published in 1998. In this study, the authors explore the relationship between a country's legal and financial systems, investigating how legal structures influence the development of financial markets and the allocation of resources. The authors assume that legal systems that protect property rights and contracts are crucial for the development of the financial market and the economy in general (Graff 2008).

Wasiuzzaman et al. (2022) apply this theory to investigate how perceptions of investment, technology and legal risks are associated with the propensity to invest in EC campaigns.

Hornuf et al. (2022) build their theoretical framework on *agency theory* (Aghion and Bolton 1992; Grossman and Hart 1983), claiming that investors who can exercise control over a company have a positive effect on the entrepreneur's incentives, leading to better business results. Finding high-quality companies, transacting with them and monitoring them is more affordable and less costly if the costs of these activities for investors are low. The authors use this theory to test whether investments with a local orientation have a greater or lesser probability of failure.

Social-psychology theories

Two theories (*the theory of planned behavior and commitment-trust theory*) are considered in both the psychological and sociological sciences (i.e., social psychology) to accurately reflect the theoretical affiliations with the respective branches.

The *theory of planned behavior* (TPB) is a learning theory proposed by Ajzen (1985, 1991), who asserts that the best predictor of an individual's behavior consists of the intention to perform the behavior. Nevertheless, the TPB also introduces other components into the determinants of behavior. Hence, Ajzen (1985) strives to complement the earlier theory of reasoned action (Fishbein 1967) and explains that the attitude toward specific behavior, subjective norms, and perceived behavioral control—the perception of the ease of performing the behavior (Ajzen 1991)—may also influence the intention to behave (Cameron et al. 2012). The theory mainly covers consumer behavior, and few studies address managerial or employee-related issues (Ulker-Demirel and Ciftci 2020). Within the analyzed sample, Rizwan and Mustafa (2022) use the TPB to test, in the context of crowdfunding in Pakistan, whether two components of trust (craft and relational trust) impact the intention to invest. Furthermore, they analyze the mediating role of these two components between platform-related factors and the intention to invest itself. The analysis results confirm direct trust and partial mediating relationships with intention.

Commitment-trust theory is widely adopted in marketing studies, as it predicts that commitment and trust directly lead to cooperative behavior that fosters successful relationship marketing (Morgan and Hunt 1994). The model presented by Morgan and Hunt (1994) uses trust and relational commitment as mediators between five antecedents (relationship termination cost, relationship benefits, shared values, communication, and opportunistic behavior) and five outcomes (acquiescence, abandonment propensity, cooperation, functional conflict, and decision uncertainty) (Mukherjee and Nath 2007). Kang et al. (2016) aim to extend trust theory by investigating the mediating effects of computational and relationship trust on lenders' willingness to invest. Although Kang et al. (2016) refer to the work of Morgan and Hunt (1994), the authors focus exclusively on the role of trust '*in any relationship: in social structures, as well as in business and interpersonal relationships*' (Kang et al. 2016), do not consider the role of commitment, and subsequently refer to a single theory of trust.

Context

Following Roy Bhattacharjee et al. (2022), the context examination considers two aspects: the project category (industry) and the geographical area of analysis.

Table 5 summarizes the categories of EC projects that are the subject of empirical investigation in the 18 sample papers. As shown in Fig. 6, even if the scope of the analysis is limited to only EC campaigns, the number of papers that do not report a specific category remains relevant (seven). However, it is now possible to identify a dominant category under analysis (nine, 50%), namely, the technology-oriented campaign. This information provides further input than does the previous SLR, specifically that of Deng et al. (2022). The technology component is prevalent in the sample and aligns with the nature of crowdfunding platforms, particularly with ECs. Excluding reward-based (e.g., Kickstarter) or donation-based crowdfunding platforms from the sample reduced the representation of categories such as fashion and art.

The interest of researchers in the topic of sustainability (four; 22.2%), which follows the trend recorded in other research fields (especially management), is also evident. Excluding the residual class ‘Others services’ (equal to ‘Sustainability’), the sample consists of the categories ‘Food and Drink’ and ‘Finance’ (two, respectively), ‘Health’, ‘Real estate’, ‘Technical Services’ and ‘Fashion and art’ (one).

From the analysis of the articles, there appears to be little enthusiasm for disseminating detailed information about the investigated category, except for Zhang et al. (2019) (in reward-based crowdfunding). Additionally, there is a perceived risk of confusion between the various categories; for instance, the term ‘FinTech’ could be associated with both technological and financial campaigns.

The article that considered the most categories (five) is Hervé et al. (2019)—Technology, Sustainability, Health, Real Estate and Other Services—followed by Mahmood et al. (2019) with four—Technology, Sustainability, Finance and Fashion and Art.

Table 5 EC project category per article in sample

Project's category	N° of articles	% of articles	Sample articles
Technology	9	50.0%	Barbi et al. (2023); Cicchiello and Kazemikhasragh (2022); Giudici et al. (2020); Goethner et al. (2021); Hervé et al. (2019); Hornuf et al. (2022); Mahmood et al. (2019); Mohammedi and Shafi (2018); Rizwan and Mustafa (2022)
Undefined	7	38.9%	Alharbey and Van Hemmen (2021); Bapna and Ganco (2021); Feola et al. (2021); Johnson et al. (2018); Kang et al. (2016); Stevenson et al. (2019b, a); Wasiuzzaman et al. (2022)
Sustainability	4	22.2%	Ciobataru et al. (2021); Hervé et al. (2019); Mahmood et al. (2019); Tenner and Hörisch (2021)
Others services	4	22.2%	Barbi et al. (2023); Goethner et al. (2021); Hervé et al. (2019); Hornuf et al. (2022)
Consumer products	2	11.1%	Goethner et al. (2021); Mahmood et al. (2019)
Food and drink	2	11.1%	Goethner et al. (2021); Mahmood et al. (2019)
Finance	2	5.6%	Hornuf et al. (2022); Mahmood et al. (2019)
Health	1	5.6%	Hervé et al. (2019)
Real Estate	1	5.6%	Hervé et al. (2019)
Technical services	1	5.6%	Hornuf et al. (2022)
Fashion and art	1	5.6%	Mahmood et al. (2019)

Table 6 presents the geographical reference areas of the analyses conducted in the sample articles. Europe has emerged as the area with the most investigations, with nine articles. Within Europe, Italy and Germany each have three studies, whereas France, Sweden, and the Netherlands are represented by one study each. Excluding the two articles that do not explicitly define an analysis population (Bapna and Ganco 2021; Mahmood et al. 2019), the USA ranks second with two articles (Johnson et al. 2018; Stevenson et al. 2019b, a), followed by all other countries with one article each (i.e., Pakistan, Malaysia, Saudi Arabia, Latin America, and China).

China is the country that has seen the most significant decrease in the share of articles compared with the entire sample; this is an indication of how the analyses in China are more focused on the lending side of crowdfunding, in line with the impressive spread that peer-to-peer lending-based platforms have had in this territory (Ding et al. 2021).

Characteristics

Table 7 shows the dependent variables that pertain to the likelihood of successful or intersecting relationships (e.g., Giudici et al. 2020) on EC platforms. The findings for these variables are consistent with those presented in Fig. 8, which indicates a diverse and varied use of terms often related to the likelihood of investment. Consequently, the SLR focuses solely on the determinants of EC project success, considering the variables that demonstrated significant coefficients in the empirical analyses.

To enhance the readability of Table 7, the independence—or control (Mahmood et al. 2019) variables are categorized into four blocks of characteristics: sociodemographic, psychological and attitudinal, competence or experience and those related to user perceptions.

Considering the number of repetitions (27), the first block is related to the sociodemographic characteristics of the EC investors. Specifically, the variables identified refer to ‘gender’ (13 repetitions), ‘age’ (eight), ‘geographical location’ (five), and ‘education’ (one). Notably, investor gender is the most frequently used aspect compared with all the blocks. The frequent use of this demographic variable aligns with the theoretical frameworks assumed in the articles. Certainly, gender is central in the evaluation of relationships

Table 6 Geographical area of analysis in the EC samples

Geographical area	N° of articles	% of articles	Sample article
Italy	3	16.7%	Barbi et al. (2023); Feola et al. (2021); Giudici et al. (2020)
Germany	3	16.7%	Goethner et al. (2021); Hornuf et al. (2022); Tenner and Hörisch (2021)
USA	2	11.1%	Johnson et al. (2018); Stevenson et al. (2019b, a)
Undefined	2	11.1%	Bapna and Ganco (2021); Mahmood et al. (2019)
France	1	5.6%	Hervé et al. (2019)
Pakistan	1	5.6%	Rizwan and Mustafa (2022)
Sweden	1	5.6%	Mohammadi and Shafi (2018)
Netherlands	1	5.6%	Ciobataru et al. (2021)
Malaysia	1	5.6%	Wasiuzzaman et al. (2022)
Saudi Arabia	1	5.6%	Alharbey and Van Hemmen (2021)
Latin america	1	5.6%	Cicchello and Kazemikhasragh (2022)
China	1	5.6%	Kang et al. (2016)

Table 7 Variables investigated in EC investors' characteristics research

Sample articles	Dependent variable	Independent variable	Relation direction	Note
Johnson et al. (2018)	Trustworthiness	Gender (Female)	+	Categories: Casual Investors, Crowd Enthusiasts, and Sophisticated Investors
	Willingness	Trustworthiness	+	
		Competence	+	
Barbi et al. (2023)	Amount invested	Age	+	
		Location (North)	+	
		Investment experience	+	
Tenner and Hörisch (2021)	Likelihood to invest	Age	–	
		Education	+	
		Familiarity with crowd-funding	+	
		Self-enhancement values	–	
Goethner et al. (2021)	Investment decision	Conservative value	–	
		Investors category		
Hervé et al. (2019)	Amount Invested	Age	+	
		Gender (Male)	+	
		Geographical proximity	+	
	Probability of a follow-up investment	Age	+	
		Gender (Male)	+	
		Geographical proximity	–	
	Funded dummy	Age	+	
		Gender (Male)	–	
	Achieved funding	Age	+	
Rizwan and Mustafa (2022)	Intention to invest	Calculus trust	+	Trust mediates the platform characteristics and intention to invest relationship
		Relational trust	+	
Mohammadi and Shafi (2018)	Firm age	Gender (Female)	+	Experience moderate Gender-Interest relationship
	Technology firm	Gender (Female)	–	
	Equity investment	Gender (Female)	–	
Bapna and Ganco (2021)	Interest in investment	Gender (Female)	+	
		Experience	–	
Ciobotaru et al. (2021)	Preference in crowd-funding	Gender (Female)	+	
	Crowdfunding knowledge	Financial literacy	+	
Wasiuzzaman et al. (2022)	Proportion of investment portfolio	Investment risk perception	–	Direct relation and mediation with familiarity
		Legal risk perception	+	
		Experience	+	
Alharbey and Van Hemmen (2021)	Intention to invest	Trust in platform	+	

Table 7 (continued)

Sample articles	Dependent variable	Independent variable	Relation direction	Note
Feola et al. (2021)	Investor clusters	Age		Investor clusters: venture trustful, crowdfunding technicians, financial investors, talent scouts and social dreamers
		Gender		
		Ease of use		
		Intrinsic motivation		
		Safety online transaction feeling		
		Likelihood to reuse crowdfunding		
Cicchiello and Kazemikhasragh (2022)	Number of male/female investors	Gender (Female entrepreneur)	+	Inverted relationship: investors characteristics are dependent variables
	Gender (Female)	Risk tolerance	–	
	Gender (Female)	Firm age	+	
Hornuf et al. (2022)	Local bias	Personal ties	+	Trust beliefs used as mediators between project, platform and fundraiser characteristics and willingness to invest
Kang et al. (2016)	Willingness to invest	Calculus trust	+	
		Relational trust	+	
Mahmood et al. (2019)	Amount invested	Experience	+	<u>Used as control</u> ; independent variable is logo complexity
Giudici et al. (2020)	Realized tie	Geographical proximity	+	Searching effort mediates self-efficacy-decision making relation
		Age (same age class)	+	
Stevenson et al. (2019b, a)	Searching effort	Self-efficacy	–	
	Decision-making performance	Searching effort	+	
	Searching effort	Knowledge	+	

inherent in homophily theory (Cicchiello and Kazemikhasragh 2022), gender role theory (Bapna and Ganco 2021), the stereotype–content model (Johnson et al. 2018), and social network theory (Hervé et al. 2019). Cicchiello and Kazemikhasragh (2022) assess the probability of having female investors on the basis of the presence of female entrepreneurs. Moreover, investor gender is employed as a determinant of risk tolerance and firm age. Hervé et al. (2019) used sociodemographic variables the most, aiming to assess their impact on various measures of campaign success (e.g., amount invested, probability of follow-up investment, achieved funding, and funding dummy). While the age of investors is commonly considered, it does not appear to be central in the analyses of the sample articles, except for Hervé et al. (2019) and Giudici et al. (2020), who also extensively employ geographical proximity, aligning with social network theory.

The block of psychological and attitudinal variables is the second most frequently utilized block (occurring 13 times) and is characterized by eight variables. This cluster significantly preponders the use of ‘trust’ (six times). This abundance is attributable to the adoption of theories such as Transfer Trust Theory (Alharbey and Van Hemmen 2021) and Commitment-Trust Theory (Kang et al. 2016). Kang et al. (2016) and Rizwan and Mustafa (2022) notably contribute to the increased frequency by distinguishing between ‘trust in calculus’ and ‘relational trust’ (with respect to Commitment-Trust Theory and

Theory of Planned Behavior, respectively). The other variables are each utilized once, reflecting the diversity of psychological and social theories. This grouping intends to include the investor category as an attitude toward belonging to a specific behavioral class (Goethner et al. 2021). Additionally, typical variables from the psychology sciences (excluding trust) include self-efficacy (Stevenson et al. 2019b, a) and self-enhancement (Tenner and Hörisch, 2021).

The ‘competence and experience’ block ranks third in the frequency of repetition (eight occurrences) and comprises four variables. ‘Experience’ is the most frequently utilized variable within the cluster, occurring five times. It serves as a direct determinant for variables indicating campaign success and as a moderator in the relationship between gender and interest in investing (Bapna and Ganco 2021). Mahmood et al. (2019) employ experience as a control variable. Following this are ‘competence’ (Johnson et al. 2018), ‘financial literacy’ (Ciobotaru et al. 2021), and ‘financial knowledge’ (Stevenson et al. 2019b, a).

The last block relates to user perceptions, the frequency of which is four, with as many variables used in two articles: ‘Ease of use’ and ‘safe online transaction feeling’ (Feola et al. 2021), ‘legal risk perception,’ and ‘investment risk perception’ (Wasiuzzaman et al. 2022).

Methods

As mentioned in the research methodology (paragraph 4), the search for articles was carried out to incorporate only quantitative analyses. Three characteristics illustrate the analysis and data collection methodologies used in the subsample of 18 articles (Table 8): methodology, tools, and analysis techniques.

Compared with Fig. 7, in the EC subsample, there is a predominance of articles (eight) that use neither survey (seven) nor experimental design (three). In this case, the designation ‘Empirical’ is used for articles that use data from datasets that are already complete and available for analysis (Hervé et al. 2019; Mahmood et al. 2019; Mohammadi and Shafi 2018) or formed through a process of online information picking, i.e., hand collection (Barbi et al. 2023; Cicchiello and Kazemikhasragh 2022; Giudici et al. 2020; Goethner et al. 2021), or a mix of the two methods (Hornuf et al. 2022). The most frequently used analysis technique in these articles is the regression model. However, Poisson regression (Cicchiello and Kazemikhasragh 2022), cluster analysis with ANOVA (Goethner et al. 2021), and probit models for dummy dependent variables (Giudici et al. 2020) are also available.

Surveys in all cases use questionnaires for data collection and employ a variety of analysis techniques, such as SEM or PLS-SEM (Alharbey and Van Hemmen 2021; Rizwan and Mustafa 2022; Wasiuzzaman et al. 2022), probit models (Ciobotaru et al., 2021) and logistic regression (Tenner and Hörisch, 2021), cluster analysis with ANOVA (Feola et al. 2021), and bootstrapping (Kang et al. 2016).

Finally, only three articles followed an experimental protocol for evaluating the hypotheses formulated (Bapna and Ganco 2021; Johnson et al. 2018; Stevenson et al. 2019b, a). In all three articles, the tools used are identified as questionnaires; in the case of Bapna and Ganco (2021), the administration was mailed. The regression model for

Table 8 Types of quantitative data collection methods and data analysis techniques

Articles	Methodology	Tool	Analysis technique
Alharbey and Van Hemmen (2021)	Survey	Questionnaire	SEM
Bapna and Ganco (2021)	Experimental design	Mailing + Questionnaire	Regression model
Barbi et al. (2023)	Empirical	Hand-collected	Regression model
Cicchello and Kazemikhasragh (2022)	Empirical	Hand-collected	Poisson regression
Ciobotaru et al. (2021)	Survey	Questionnaire	Probit model
Feola et al. (2021)	Survey	Questionnaire	Cluster analysis + ANOVA
Giudici et al. (2020)	Empirical	Hand-collected	Probit model
Goethner et al. (2021)	Empirical	Hand-collected	Cluster analysis + ANOVA
Hervé et al. (2019)	Empirical	Database	Regression model
Hornuf et al. (2022)	Empirical	Database + Hand-collected	Regression model
Johnson et al. (2018)	Experimental design	Questionnaire	Path analysis
Kang et al. (2016)	Survey	Questionnaire	Bootstrap
Mahmood et al. (2019)	Empirical	Database	Regression model
Mohammadi and Shafi (2018)	Empirical	Database	Regression model
Rizwan and Mustafa (2022)	Survey	Questionnaire	PLS-SEM
Stevenson et al. (2019b, a)	Experimental design	Questionnaire	Regression model + Chow test
Tenner and Hörisch (2021)	Survey	Questionnaire	Logistic regression
Wasiuzzaman et al. (2022)	Survey	Questionnaire	PLS-SEM

Bapna and Ganco (2021) and Stevenson et al. (2019b, a) (with a Chow test) and the path analysis of Johnson et al. (2018) are the analysis techniques implemented.

Notably, some articles (e.g., Stevenson et al. 2019b, a) developed multiple studies in the same paper; however, in the present SLR, only the quantitative part, which refers to the characteristics of the funders (investors for the subsample), was taken into account.

The paucity of studies with an experimental design could be attributed to the higher cost and time that such a procedure requires. Stevenson et al. (2019b, a) adopt an experimental methodology in line with their focus on perceptual-control theory, which is grounded in experimental psychology.

Interconnections between TCCM components in the EC context

In this section, we emphasize the relationships among the components of the TCCM framework, revealing significant interconnections that enhance the understanding of EC dynamics and investment decisions. First, there is strong interdependence between theories and context. For example, social network theory, often applied to technology projects (Hervé et al. 2019), emphasizes how social networks and relational dynamics are crucial for the success of innovative projects. Similarly, the theory of basic human values is closely associated with sustainability projects (Tenner and Hörisch, 2021), suggesting that personal values such as self-enhancement and conservatism change influence investment decisions in green initiatives. This link between Theory and Context is further supported by geographical evidence: studies conducted in Europe (Italy, Germany, and France) highlight the critical role of geographical proximity as a determining factor in investment decisions (Giudici et al. 2020; Hervé et al. 2019), whereas in non-European

contexts, such as the USA or Saudi Arabia, the focus shifts to cultural dynamics and trust in platforms (Alharbey and Van Hemmen 2021).

Gender, as a central characteristic, emerges transversally across theories and contexts. Gender role theory (Bapna and Ganco 2021) and the stereotype content model (Johnson et al. 2018) explore how gender roles and stereotypes influence perceptions and decisions. For instance, female gender is positively associated with trust in crowdfunding platforms (Johnson et al. 2018) and a preference for socially responsible projects (Cio-botaru et al., 2021). However, the influence of gender varies across contexts and sectors, as revealed by studies on technology projects, where men show a greater propensity to invest in high-tech sectors (Mohammadi and Shafi 2018), and sustainability projects, where women demonstrate greater sensitivity to environmental values (Tenner and Hörisch, 2021).

To deepen this multidimensionality, studies such as those by Rizwan and Mustafa (2022) and Alharbey and Van Hemmen (2021) show that trust, a key component in investment behavior, is shaped by demographic analysis. For example, studies examining how gender differences influence trust in crowdfunding platforms combine psychological theories, such as the stereotype content model (Johnson et al. 2018), with demographic data to analyze how personal, cultural, and social factors intersect to determine investment intentions. These multidimensional approaches are particularly useful in explaining the complexity of causal relationships, such as the role of calculated and relational trust in equity-based crowdfunding platforms (Rizwan and Mustafa 2022).

Finally, trust emerges as a cross-cutting concept tied to theoretical and contextual dimensions. Theories such as Swift Trust Theory (Alharbey and Van Hemmen 2021) and Commitment-Trust Theory (Kang et al. 2016) demonstrate that the success of crowdfunding projects depends on a balance between immediate trust and long-term relational building.

Summary of findings and future research directions

One of the key objectives of this SLR is to highlight the importance of incorporating investors' perspectives and characteristics in EC campaigns. This section addresses RQ4 by utilizing the TCCM framework to guide future research directions. By summarizing the main findings, the framework serves as a structured approach to identify and prioritize areas for further investigation.

Theories

The results strongly emphasize the use of psychological theories to analyze the characteristics of EC investors. Among the 18 articles reviewed, 13 distinct theoretical frameworks were identified, with a notable trend toward incorporating multiple theories within individual studies. The theories span psychology, sociology, economics, and social psychology. Psychological frameworks, including Trust Transfer Theory and Gender Role Theory, dominate the literature, exploring dynamics such as trust and gender biases in investment behavior. Sociological theories such as homophily theory and social network theory highlight the importance of social relationships and shared characteristics between investors and entrepreneurs. Economic theories, including law and finance theory, examine the role of legal and financial structures

in investment decisions, whereas social–psychological theories such as the theory of planned behavior emphasize the interplay of intention and trust.

While the diversity of theories used in the literature enriches the understanding of investor behavior, it also raises concerns about theoretical fragmentation. This lack of cohesion can hinder the establishment of a unified perspective on the factors influencing investment decisions in ECs. The preponderance of psychological theories suggests a need for a more integrated framework that combines both bottom-up and top-down approaches. This would entail considering the stimuli that investors encounter alongside their cognitive responses, as well as demographic factors that may influence these interactions. The isolated attempt by Wan Mohamad Nazarie and Williams (2021) to merge signaling theory with homophily theory demonstrates the potential for interdisciplinary approaches but also highlights the limited application of such integrations to broader contexts beyond reward-based crowdfunding.

To advance the field, future studies should aim to develop comprehensive theoretical frameworks that encompass psychological, sociological, and economic theories and simultaneously address the role played by visual stimuli displayed to investors, cognitive functions, and demographic characteristics. The incorporation of economic theories, as suggested by Hervé et al. (2019), could provide valuable insights into the evolving sophistication of investors within the crowdfunding landscape. Furthermore, given the growing relevance of consumer neuroscience, integrating neuroeconomic frameworks may enhance the understanding of decision-making processes in the EC. This multidimensional approach could ultimately yield a more coherent and actionable understanding of investor behavior, facilitating more effective strategies for crowdfunding.

Context

The examination of context in the EC literature shows a dominant focus on technology-oriented campaigns, which account for 50% of the sample. Sustainability appears in four articles, but other categories, such as food and drink and finance, receive less attention. The potential for confusion in categorization, particularly with terms such as “FinTech”, highlights the need for clearer definitions.

While the dominance of technology in the EC literature underscores its relevance to current crowdfunding practices, the lack of specificity in many papers limits the potential for nuanced insights. The preponderance of unspecified categories could hinder a comprehensive understanding of the dynamics within different sectors, making it difficult for researchers and practitioners to apply findings effectively. Additionally, the underrepresentation of sustainability suggests a gap in addressing emerging trends. The issues of category overlap, particularly in areas such as fintech and sustainable projects, reflect the complexity of modern crowdfunding landscapes, indicating an urgent need for clearer frameworks and definitions to enhance research rigor and applicability.

Future studies should include more geographical diversity, particularly in developing countries, to explore the unique contributions of EC platforms. Expanding the analysis of project categories and establishing clearer definitions will enhance research applicability and provide a more comprehensive understanding of the evolving EC landscape.

Characteristics

The review identifies four main blocks of variables influencing EC success: sociodemographic characteristics, with a strong emphasis on investor gender; psychological and attitudinal factors, particularly trust, competence, and experience; and user perception variables, such as ease of use and risk perception. The frequent focus on gender aligns with theories such as homophily theory, highlighting its significance in understanding investor behavior.

The strong focus on gender underscores its importance in EC research but may limit the exploration of other relevant factors. While sociodemographic aspects provide valuable insights, the limited engagement with psychological traits and cognitive dimensions suggests a need for a more holistic approach. The interplay between demographic factors and psychological variables is crucial for understanding investor behavior comprehensively, yet much of the current research remains fragmented.

Future research should explore gender by incorporating investors' cognitive and psychological factors. This approach could help create a comprehensive framework that defines the psychological traits of EC investors, forming a foundation for a bottom-up understanding of how they respond to platform stimuli. Additionally, it would be valuable to examine the financial awareness of EC investors, as they are becoming more similar to traditional investors.

Methods

The findings reveal a predominance of empirical analyses over surveys and experimental designs. Many articles employed preexisting datasets or collected data through online methods, with regression models being the most common analysis technique. Only a few studies have utilized experimental protocols, which could be due to the higher costs and time demands associated with such approaches. In any case, purely experimental research has already highlighted the inadequacy of data obtained through traditional tools in investigating customer decision-making behavior (Calvert and Brammer 2012; Hsu and Cheng 2018), whose data are considered limited or '*insincere*' (Nilashi et al. 2020). One way to overcome some limitations is to include neuroscientific techniques in the data collection phase, which can also identify individuals' automatic and unconscious activity in the financial decision process (Camerer et al. 2005; Miendlarzewska et al. 2019; Plassmann et al. 2012). Such a study setting not only preserves the natural decision-making processes but also enables researchers to quantify the weight and relevance of information provided on the EC platform and campaigns, matching bottom-up (signal sets) and top-down approaches (the functions of the cognitive structures of investors) (i.e., different theories). In particular, the visual attention of financial investors is typically intercepted through eye tracking, which has enabled researchers to develop a new tool that can track the eye movements of individuals, linking what is in the visual field and the information that the brain processes ('eye-mind hypothesis'—Just and Carpenter 1980). This type of tool has already been applied in studies related to consumer behavior, both in the marketing field (Husić-Mehmedovic et al., 2017; Kim et al. 2018; Menon et al. 2016; Wolfson et al. 2017) and in the financial context regarding investment choices and biases (Agnoli et al. 2015; Duclos 2015; Hüsser and Wirth 2014).

Conclusions

In recent years, crowdfunding has become a popular alternative to traditional financing methods, streamlining access to credit and enabling quicker equity subscriptions for innovative start-ups. This shift has attracted both global investors and retail lenders, making it essential to understand the factors that drive success in this domain. A significant body of literature has focused on the characteristics of crowdfunding pages and campaigns to identify key predictors of success. This paper aims to conduct an SLR on the role of funders' characteristics, recognizing that individual traits play a crucial role in shaping investor behavior.

Over the past eight years, research on funders' characteristics in crowdfunding has expanded considerably, with a growing focus on individual differences among investors. Most studies are published in high-impact journals and primarily emphasize economics, business, and management. Given the crucial role that the EC plays in driving financial innovation, we applied the TCCM framework to review the literature on this specific subsample of articles. Our analysis identifies key gaps and underexplored topics, offering valuable insights that clarify the current state of research on funder characteristics in ECs. Notably, while there is a strong focus on psychology-related theories, no single dominant theory emerges, although gender-based theories show some prevalence. Consistent with the broader crowdfunding literature, EC research has focused predominantly on technology and sustainability campaigns, particularly within European contexts. Our findings indicate that many dependent variables, such as the amount invested, are closely tied to campaign success, whereas independent variables often involve demographic traits—especially gender—and psychological factors, with trust being a prominent theme. However, a notable gap is the lack of empirical evidence from experimental methods, as most studies rely on databases (online or hand-collected) and surveys.

To further advance the field, we suggest several avenues for future research. First, a comprehensive theoretical framework that integrates various investor characteristics—including demographic, psychological, and skill-based factors—is necessary. Additionally, employing more experimental methodologies, supported by neuroscientific techniques, could uncover automatic and unconscious decision-making processes that traditional approaches often miss. The current literature, which is primarily grounded in management, economics, and business, tends to neglect psychological and neuroscientific perspectives. This gap limits our understanding of the cognitive and emotional factors that influence investor behavior, which could lead to superficial analyses of funding motivations. Moreover, the reliance on surveys, rather than experimental designs, restricts insights into the complexities of decision-making. An overemphasis on sociodemographic factors can also obscure other critical influences, such as motivations and emotional engagement. While the theoretical landscape is diverse, this variety can lead to fragmentation, hindering a unified understanding of funder behavior. A more multidimensional approach, therefore, will deepen our understanding of investor behavior and contribute to the development of more effective EC strategies.

Like previous articles, this SLR is not without its limitations. One key limitation is the absence of a third data provider, such as Scopus, alongside EBSCOhost and Web of Science. Additionally, the constraints imposed by the limited number of articles reviewed restrict the scope of our research, underscoring the need for future studies to investigate

a broader range of literature and contexts to enhance understanding in this area. The current scarcity of articles on ECs, coupled with the diversity of theories explored, makes it premature to establish a comprehensive framework. However, this challenge offers opportunities for future SLRs as the literature continues to evolve. These limitations also indicate several directions for future research. First, it could benefit from incorporating gray literature, books, and chapters to provide a more holistic view of funder characteristics in equity-based crowdfunding. Second, a more nuanced examination of how investor characteristics interact with campaign features could enrich the literature and yield valuable insights. Subsequent studies could build on our findings to gain intriguing insights into the interplay between investor characteristics and campaign features.

Author contributions

All author(s) read and approved the final manuscript.

Funding

Not applicable.

Availability of data and materials

Not applicable.

Declarations

Competing interests

The author declare that they have no competing interests.

Received: 12 July 2024 Accepted: 11 October 2025

Published online: 06 January 2026

SLR sample references

- Ahn S (2023) To buy or to back? Backers' motivations for fashion crowdfunding projects. *J Fashion Mark Manag* 27(2):335–351
- Alharbey M, Van Hemmen S (2021) Investor intention in equity crowdfunding. Does trust matter? *J Risk Financ Manag* 14(2):53
- Arifah JN, Dalimunthe Z (2020) The impact of financial literacy on the investment decision of non-donation-based crowdfunding in Indonesia. *Int J Bus Soc* 21(3):1045–1057
- Bapna S, Ganco M (2021) Gender gaps in equity crowdfunding: evidence from a randomized field experiment. *Manag Sci* 67(5):2679–2710
- Barbi M, Febo V, Giudici G (2023) Community-level social capital and investment decisions in equity crowdfunding. *Small Bus Econ* 61(3):1–36
- Cecere G, Le Guel F, Rochelandet F (2017) Crowdfunding and social influence: an empirical investigation. *Appl Econ* 49(57):5802–5813
- Chen D, Lou H, Van Slyke C (2015) Toward an understanding of online lending intentions: evidence from a survey in China. *Commun Assoc Inf Syst* 36(1):17
- Chen J, Jiang J, Liu YJ (2018) Financial literacy and gender difference in loan performance. *J Empir Finance* 48:307–320
- Cicchello AFF, Kazemikhasragh A (2022) Tackling gender bias in equity crowdfunding: an exploratory study of investment behaviour of Latin American investors. *Eur Bus Rev* 34(3):370–395
- Ciobotaru LC, Kim S, van Soest A (2021) Household preferences for investing in crowdfunding. *De Economist* 169:499–522
- Davis BC, Hmieleski KM, Webb JW, Coombs JE (2017) Funders' positive affective reactions to entrepreneurs' crowdfunding pitches: the influence of perceived product creativity and entrepreneurial passion. *J Bus Ventur* 32(1):90–106
- Feola R, Vesci M, Marinato E, Parente R (2021) Segmenting "digital investors": evidence from the Italian equity crowdfunding market. *Small Bus Econ* 56:1235–1250
- Giudici G, Guerini M, Rossi-Lamastra C (2020) Elective affinities: exploring the matching between entrepreneurs and investors in equity crowdfunding. *Balt J Manag* 15(2):183–198
- Goethner M, Luettig S, Regner T (2021) Crowdfunding in entrepreneurial projects: disentangling patterns of investor behavior. *Small Bus Econ* 57:905–926
- Greenberg J, Mollick E (2017) Activist choice homophily and the crowdfunding of female founders. *Adm Sci Q* 62(2):341–374
- Groza MP, Groza MD, Barral LM (2020) Women backing women: the role of crowdfunding in empowering female consumer-investors and entrepreneurs. *J Bus Res* 117:432–442

- Harkness SK (2016) Discrimination in lending markets: status and the intersections of gender and race. *Soc Psychol Q* 79(1):81–93
- Hervé F, Manthé E, Sannajust A, Schwienbacher A (2019) Determinants of individual investment decisions in investment-based crowdfunding. *J Bus Finance Account* 46(5–6):762–783
- Hong J, Byun KA (2020) Attracting prosocial lenders from different cultures to help others in microlending. *J Consum Mark* 37(2):205–214
- Hornuf L, Schmitt M, Stenzhorn E (2022) The local bias in equity crowdfunding: behavioral anomaly or rational preference? *J Econ Manag Strategy* 31(3):693–733
- Johnson MA, Stevenson RM, Letwin CR (2018) A woman's place is in the... startup! Crowdfunder judgments, implicit bias, and the stereotype content model. *J Bus Venturing* 33(6):813–831
- Kang M, Gao Y, Wang T, Zheng H (2016) Understanding the determinants of funders' investment intentions on crowdfunding platforms: a trust-based perspective. *Ind Manag Data Syst* 116(8):1800–1819
- Liang TP, Wu SPJ, Huang CC (2019) Why funders invest in crowdfunding projects: role of trust from the dual-process perspective. *Inf Manag* 56(1):70–84
- Liang X, Hu X, Li EY, Meng H (2023) Untangling the influence of perceived sustainability orientation on value-co-creation behavior in crowdfunding process: investigating a mediation model. *Internet Res*. <https://doi.org/10.1108/INTR-12-2021-0921>
- Mahmood A, Luffarelli J, Mukesh M (2019) What's in a logo? The impact of complex visual cues in equity crowdfunding. *J Bus Venturing* 34(1):41–62
- Mohammadi A, Shafi K (2018) Gender differences in the contribution patterns of equity-crowdfunding investors. *Small Bus Econ* 50:275–287
- Moysidou K, Hausberg JP (2020) In crowdfunding we trust: a trust-building model in lending crowdfunding. *J Small Bus Manag* 58(3):511–543
- Rizwan A, Mustafa F (2022) Fintech attaining sustainable development: an investor perspective of crowdfunding platforms in a developing country. *Sustainability* 14(12):7114
- Shneor R, Munim ZH (2019) Reward crowdfunding contribution as planned behaviour: an extended framework. *J Bus Res* 103:56–70
- Stevenson RM, Ciuchta MP, Letwin C, Dinger JM, Vancouver JB (2019a) Out of control or right on the money? Funder self-efficacy and crowd bias in equity crowdfunding. *J Bus Venturing* 34(2):348–367
- Sutanto J, Wenninger H, Duriana H (2021) Warm-glow giving, hedonism, and their influence on Muslim user engagement on loan-based crowdfunding platforms. *J Assoc Inf Syst* 22(2):7
- Tenner I, Hörisch J (2021) Crowdfunding sustainable entrepreneurship: What are the characteristics of crowdfunding investors? *J Clean Prod* 290:125667
- Vaznyte E, Andries P, Manigart S (2023) Are you part of the crowd? The role of sex and environmental characteristics for crowdfunding awareness. *J Small Bus Manage* 61(2):1062–1094
- Wan Mohamad Nazarie WNF, Williams R (2021) Linguistic style and gender match in funding intention towards crowdfunding project. *Rev Int Bus Strategy* 31(3):438–461
- Wang Z, Yang X (2019) Understanding backers' funding intention in reward crowdfunding: an elaboration likelihood perspective. *Technol Soc* 58:101149
- Wasiuzzaman S, Chong LL, Ong HB (2022) Influence of perceived risks on the decision to invest in equity crowdfunding: a study of Malaysian investors. *J Entrep Emerg Econ* 14(2):208–230
- Yadav A, Mathur HP (2023) Reconnoitering antecedents of donation intention in donation crowdfunding campaigns: a mediating role of crowdfunding readiness. *Int Rev Public Nonprofit Mark* 21(1):1–26
- Yang X, Zhao K, Tao X, Shiu E (2019) Developing and validating a theory-based model of crowdfunding investment intention—perspectives from social exchange theory and customer value perspective. *Sustainability* 11(9):2525
- Zhang C, Campana PE, Liu C, Wang K, Yan J (2019) Crowdfunding preferences for a sustainable milk product with integrated photovoltaic water pumping system in China. *Appl Energy* 255:113694
- Zhang F, Zhang H, Gupta S (2023) Investor participation in reward-based crowdfunding: impacts of entrepreneur efforts, platform characteristics, and perceived value. *Inf Technol Manag* 24(1):19–36
- Zhang H, Chen W (2019) Backer motivation in crowdfunding new product ideas: is it about you or is it about me? *J Prod Innov Manag* 36(2):241–262
- Zhao Q, Chen CD, Wang JL, Chen PC (2017) Determinants of backers' funding intention in crowdfunding: social exchange theory and regulatory focus. *Telemat Inform* 34(1):370–384

References

- Aghion P, Bolton P (1992) An incomplete contracts approach to financial contracting. *Rev Econ Stud* 59(3):473–494. <https://doi.org/10.2307/2297860>
- Agnoli S, Franchin L, Rubaltelli E, Corazza GE (2015) An eye-tracking analysis of irrelevance processing as moderator of openness and creative performance. *Creat Res J* 27(2):125–132. <https://doi.org/10.1080/10400419.2015.1030304>
- Ahsan M, Cornelis EF, Baker A (2018) Understanding backers' interactions with crowdfunding campaigns: co-innovators or consumers? *J Res Mark Entrep* 20(2):252–272. <https://doi.org/10.1108/JRME-12-2016-0053>
- Ajzen I (1985) From intentions to actions: a theory of planned behavior. *Action control: from cognition to behavior*. Springer, Berlin Heidelberg, pp 11–39. https://doi.org/10.1007/978-3-642-69746-3_2
- Ajzen I (1991) The theory of planned behavior. *Organ Behav Hum Decis Process* 50(2):179–211. [https://doi.org/10.1016/0749-5978\(91\)90020-T](https://doi.org/10.1016/0749-5978(91)90020-T)
- Alvino L, Pavone L, Abhishta A, Robben H (2020) Picking your brains: where and how neuroscience tools can enhance marketing research. *Front Neurosci* 14:577666. <https://doi.org/10.3389/fnins.2020.577666>
- Anderson JR (1982) Acquisition of cognitive skill. *Psychol Rev* 89(4):369. <https://doi.org/10.1037/0033-295X.89.4.369>
- Barber BM, Odean T (2001) Boys will be boys: gender, overconfidence, and common stock investment. *Q J Econ* 116(1):261–292. <https://doi.org/10.1162/003355301556400>

- Belleflamme P, Lambert T, Schwienbacher A (2014) Crowdfunding: tapping the right crowd. *J Bus Venturing* 29(5):585–609. <https://doi.org/10.1016/j.jbusvent.2013.07.003>
- Brannon DL, Deb P, Sreekumar V (2022) Crowdfunding and governance: a review and research roadmap. *Glob J Account Financ (GJAF)*, 6(2)
- Bruton G, Khavul S, Siegel D, Wright M (2015) New financial alternatives in seeding entrepreneurship: microfinance, crowdfunding, and peer-to-peer innovations. *Entrep Theory Pract* 39(1):9–26. <https://doi.org/10.1111/etap.12143>
- Burks SV, Carpenter JP, Goette L, Rustichini A (2009) Cognitive skills affect economic preferences, strategic behavior, and job attachment. *Proc Natl Acad Sci U S A* 106(19):7745–7750. <https://doi.org/10.1073/pnas.0812360106>
- Calvert GA, Brammer MJ (2012) Predicting consumer behavior: using novel mind-reading approaches. *IEEE Pulse* 3(3):38–41. <https://doi.org/10.1109/MPUL.2012.2189167>
- Camerer C, Loewenstein G, Prelec D (2005) Neuroeconomics: how neuroscience can inform economics. *J Econ Lit* 43(1):9–64. <https://doi.org/10.1257/0022051053737843>
- Cameron R, Ginsburg H, Westhoff M, Mendez RV (2012) Ajzen's theory of planned behavior and social media use by college students. *Am J Psychol Res* 8(1):1–20. <https://doi.org/10.25115/psy.v8i1.456>
- Chaney D (2019) A principal-agent perspective on consumer co-production: crowdfunding and the redefinition of consumer power. *Technol Forecast Soc Change* 141:74–84. <https://doi.org/10.1016/j.techfore.2018.06.013>
- Collins L, Pierrakis Y (2012) The venture crowd: crowdfunding equity investments into business
- Danese P, Manfè V, Romano P (2018) A systematic literature review on recent lean research: state-of-the-art and future directions. *Int J Manag Rev* 20(2):579–605. <https://doi.org/10.1111/ijmr.12156>
- Deng L, Ye Q, Xu D, Sun W, Jiang G (2022) A literature review and integrated framework for the determinants of crowdfunding success. *Financ Innov* 8(1):41. <https://doi.org/10.1186/s40854-022-00345-6>
- Ding C, Kavuri AS, Milne A (2021) Lessons from the rise and fall of Chinese peer-to-peer lending. *J Bank Regul* 22:133–143. <https://doi.org/10.1057/s41261-020-00132-2>
- Duclos R (2015) The psychology of investment behavior: (De) biasing financial decision-making one graph at a time. *J Consum Psychol* 25(2):317–325. <https://doi.org/10.1016/j.jcps.2014.11.005>
- Fishbein ME (ed) (1967) Readings in attitude theory and measurement. Wiley
- Fiske ST, Cuddy AJC, Glick P, Xu J (2002) A model of (often mixed) stereotype content: competence and warmth respectively follow from perceived status and competition. *J Pers Soc Psychol* 82(6):878–902. <https://doi.org/10.1037/0022-3514.82.6.878>
- Gigerenzer G, Goldstein DG (1996) Reasoning the fast and frugal way: models of bounded rationality. *Psychol Rev* 103(4):650. <https://doi.org/10.1037/0033-295X.103.4.650>
- Gopalakrishnan S, Ganeshkumar P (2013) Systematic reviews and meta-analysis: understanding the best evidence in primary healthcare. *J Fam Med Prim Care* 2(1):9. <https://doi.org/10.4103/2249-4863.109934>
- Graff M (2008) Law and finance: common law and civil law countries compared—an empirical critique. *Economica* 75(297):60–83. <https://doi.org/10.1111/j.1468-0335.2007.00596.x>
- Grossman SJ, Hart OD (1983) Implicit contracts under asymmetric information. *Q J Econ*. <https://doi.org/10.2307/1885377>
- Hanif MA, Santosa PB (2023) TAM Construct, trust, and religiosity for decision of Muslim lenders to use funding services on sharia peer-to-peer lending platforms (website and apps). *J Ekon Syariah Teori Dan Terapan*. <https://doi.org/10.20473/vol10iss20232pp151-168>
- Hapsari MJ, Bin Mohd Thas Thaker MA, Mohammed MO, Duasa J (2022) A qualitative investigation into crowdfunding framework as a source of financing for waqf land development. *J Islam Account Bus Res* 13(3):425–443. <https://doi.org/10.1108/JIABR-10-2020-0311>
- Heilman ME (2012) Gender stereotypes and workplace bias. *Res Organ Behav* 32:113–135. <https://doi.org/10.1016/j.riob.2012.11.003>
- Hemer J, Schneider U, Dornbusch F, Frey S, Dütschke E, Bradke C (2011) Crowdfunding und andere Formen informeller Mikrofinanzierung in der Projekt- und Innovationsfinanzierung
- Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (eds) (2019) Cochrane handbook for systematic reviews of interventions. Wiley. <https://doi.org/10.1002/9781119536604>
- Hoegen A, Steininger DM, Veit D (2018) How do investors decide? An interdisciplinary review of decision-making in crowdfunding. *Electron Mark* 28(3):339–365. <https://doi.org/10.1007/s12525-017-0269-y>
- Hsu MYT, Cheng JMS (2018) fMRI neuromarketing and consumer learning theory: word-of-mouth effectiveness after product harm crisis. *Eur J Mark* 52(1/2):199–223. <https://doi.org/10.1108/EJM-12-2016-0866>
- Husić-Mehmedović M, Omeragić I, Batagelj Z, Kolar T (2017) Seeing is not necessarily liking: advancing research on package design with eye-tracking. *J Bus Res* 80:145–154. <https://doi.org/10.1016/j.jbusres.2017.04.019>
- Hüsler A, Wirth W (2014) Do investors show an attentional bias toward past performance? An eye-tracking experiment on visual attention to mutual fund disclosures in simplified fund prospectuses. *J Financ Serv Mark* 19(3):169–185. <https://doi.org/10.1057/fsm.2014.20>
- Josefy M, Dean TJ, Albert LS, Fitza MA (2017) The role of community in crowdfunding success: evidence on cultural attributes in funding campaigns to “save the local theater.” *Entrep Theory Pract* 41(2):161–182. <https://doi.org/10.1111/etap.12263>
- Just MA, Carpenter PA (1980) A theory of reading: from eye fixations to comprehension. *Psychol Rev* 87(4):329. <https://doi.org/10.1037/0033-295X.87.4.329>
- Kim E, Tang LR, Meusel C, Gupta M (2018) Optimization of menu-labeling formats to drive healthy dining: an eye tracking study. *Int J Hosp Manag* 70:37–48. <https://doi.org/10.1016/j.ijhm.2017.10.020>
- Kumar S, Goyal N (2016) Evidence on rationality and behavioural biases in investment decision making. *Qual Res Financ Mark* 8(4):270–287. <https://doi.org/10.1108/QRFM-05-2016-0016>
- Li X, Tang Y, Yang N, Ren R, Zheng H, Zhou H (2016) The value of information disclosure and lead investor in equity-based crowdfunding: an exploratory empirical study. *Nankai Bus Rev Int* 7(3):301–321. <https://doi.org/10.1108/NBRI-01-2016-0002>

- Liu D, Ainsworth SE, Baumeister RF (2016) A meta-analysis of social networking online and social capital. *Rev General Psychol* 20(4):369–391.
- Lukkarinen A, Wallenius J, Seppälä T (2019) Investor motivations and decision criteria in equity crowdfunding. Available at SSRN 3263434
- Maehle N (2020) Sustainable crowdfunding: insights from the project perspective. *Balt J Manage* 15(2):281–302. <https://doi.org/10.1108/BJM-02-2019-0079>
- Manocha S, Bhullar PS, Sachdeva T (2023) Factors determining the investment behaviour of farmers-the moderating role of socioeconomic demographics. *J Indian Bus Res*. <https://doi.org/10.1108/JIBR-02-2022-0045>
- Mazzocchi FJ, Lucarelli C (2023) Success or failure in equity crowdfunding? A systematic literature review and research perspectives. *Manag Res Rev* 46(6):790–831. <https://doi.org/10.1108/MRR-09-2021-0672>
- McKenny AF, Allison TH, Ketchen DJ Jr, Short JC, Ireland RD (2017) How should crowdfunding research evolve? A survey of the entrepreneurship theory and practice editorial board. *Entrepr Theory Pract* 41(2):291–304. <https://doi.org/10.1111/etap.12269>
- Mendes FF, Mendes E, Salleh N (2019) The relationship between personality and decision-making: a systematic literature review. *Inf Softw Technol* 111:50–71. <https://doi.org/10.1016/j.infsof.2019.03.010>
- Menon RV, Sigurdsson V, Larsen NM, Fagerstrøm A, Foxall GR (2016) Consumer attention to price in social commerce: eye tracking patterns in retail clothing. *J Bus Res* 69(11):5008–5013. <https://doi.org/10.1016/j.jbusres.2016.04.072>
- Meyerson D, Weick KE, Kramer RM (1996) Swift trust and temporary groups. *Trust Organ: Front Theory Res* 166:195. <https://doi.org/10.4135/9781452243610.n9>
- Miendlarzewska EA, Kometer M, Preuschoff K (2019) Neurofinance. *Organ Res Methods* 22(1):196–222. <https://doi.org/10.1177/1094428117730891>
- Mollick E (2014) The dynamics of crowdfunding: an exploratory study. *J Bus Venturing* 29(1):1–16. <https://doi.org/10.1016/j.jbusvent.2013.06.005>
- Morgan RM, Hunt SD (1994) The commitment-trust theory of relationship marketing. *J Mark* 58(3):20–38. <https://doi.org/10.1177/002224299405800302>
- Mukherjee A, Nath P (2007) Role of electronic trust in online retailing: a re-examination of the commitment-trust theory. *Eur J Mark* 41(9/10):1173–1202. <https://doi.org/10.1108/03090560710773390>
- Munn Z, Peters MD, Stern C, Tufanaru C, McArthur A, Aromataris E (2018) Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. *BMC Med Res Methodol* 18(1):1–7. <https://doi.org/10.1186/s12874-018-0611-x>
- Neuhaus J, Isaak A, Bostandzic D (2022) Million dollar personality: a systematic literature review on personality in crowdfunding. *Manag Rev Q* 72(2):309–345. <https://doi.org/10.1007/s11301-021-00242-9>
- Nilashi M, Yadegaridehkordi E, Samad S, Mardani A, Ahani A, Aljojo N, Razali NS, Tajuddin T (2020) Decision to adopt neuromarketing techniques for sustainable product marketing: a fuzzy decision-making approach. *Symmetry* 12(2):305. <https://doi.org/10.3390/sym12020305>
- Olufolaji A, Phillips RA (2015) A qualitative analysis of successful crowdfunding campaigns for UK-based ventures. *IUP J Entrep Dev* 12(4):59
- Paul J, Criado AR (2020) The art of writing literature review: what do we know and what do we need to know? *Int Bus Rev* 29(4):101717. <https://doi.org/10.1016/j.ibusrev.2020.101717>
- Paul J, Rosado-Serrano A (2019) Gradual internationalization vs born-global/international new venture models: a review and research agenda. *Int Mark Rev* 36(6):830–858. <https://doi.org/10.1108/IMR-10-2018-0280>
- Paul J, Khatri P, Kaur Duggal H (2023) Frameworks for developing impactful systematic literature reviews and theory building: What, why and how? *J Decis Syst*. <https://doi.org/10.1080/12460125.2023.2197700>
- Paul J, Lim WM, O'Cass A, Hao AW, Bresciani S (2021) Scientific procedures and rationales for systematic literature reviews (SPAR-4-SLR). *Int J Consum Stud* 45(4):O1–O16. <https://doi.org/10.1111/ijcs.12695>
- Plassmann H, Ramsøy TZ, Milosavljevic M (2012) Branding the brain: a critical review and outlook. *J Consum Psychol* 22(1):18–36. <https://doi.org/10.1016/j.jcps.2011.11.010>
- Popescu D, Radu LD, Păvăloaia VD, Georgescu MR (2020) Psychological determinants of investor motivation in social media-based crowdfunding projects: a systematic review. *Front Psychol* 11:588121. <https://doi.org/10.3389/fpsyg.2020.588121>
- Porta RL, Lopez-de-Silanes F, Shleifer A, Vishny RW (1998) Law and finance. *J Polit Econ* 106(6):1113–1155. <https://doi.org/10.1086/250042>
- Powers (1973). Behavior: the control of perception
- Reber R, Schwarz N, Winkielman P (2004) Processing fluency and aesthetic pleasure: is beauty in the perceiver's processing experience? *Pers Soc Psychol Rev* 8(4):364–382. https://doi.org/10.1207/s15327957pspr0804_3
- Robb AM, Robinson DT (2014) The capital structure decisions of new firms. *Rev Financ Stud* 27(1):153–179. <https://doi.org/10.1093/rfs/hhs072>
- Roy Bhattacharjee D, Pradhan D, Swani K (2022) Brand communities: a literature review and future research agendas using TCCM approach. *Int J Consum Stud* 46(1):3–28. <https://doi.org/10.1111/ijcs.12758>
- Ruef M, Aldrich HE, Carter NM (2003) The structure of founding teams: homophily, strong ties, and isolation among US entrepreneurs. *Am Sociol Rev*. <https://doi.org/10.1177/000312240306800202>
- Rustichini A, DeYoung CG, Anderson JE, Burks SV (2016) Toward the integration of personality theory and decision theory in explaining economic behavior: an experimental investigation. *J Behav Exp Econ* 64:122–137. <https://doi.org/10.1016/j.socec.2016.04.019>
- Schwartz SH (1992) Universals in the content and structure of values: theoretical advances and empirical tests in 20 countries. *Adv Exp Soc Psychol* 25:1–65. [https://doi.org/10.1016/S0065-2601\(08\)60281-6](https://doi.org/10.1016/S0065-2601(08)60281-6)
- Schwartz SH (1994) Are there universal aspects in the structure and contents of human values? *J Soc Issues* 50(4):19–45. <https://doi.org/10.1111/j.1540-4560.1994.tb01196.x>
- Shao CY, Baker JA, Wagner J (2004) The effects of appropriateness of service contact personnel dress on customer expectations of service quality and purchase intention: The moderating influences of involvement and gender. *J Bus Res* 57(10):1164–1176. [https://doi.org/10.1016/S0148-2963\(02\)00326-0](https://doi.org/10.1016/S0148-2963(02)00326-0)

- Stanko MA, Henard DH (2016) How crowdfunding influences innovation. *MIT Sloan Manag Rev* 57(3):15
- Statista (2023). Digital Capital Raising: market data and analysis. Statista Market Insights
- Stevenson RM, Ciuchta MP, Letwin C, Dinger JM, Vancouver JB (2019b) Out of control or right on the money? Funder self-efficacy and crowd bias in equity crowdfunding. *J Bus Venturing* 34(2):348–367. <https://doi.org/10.1016/j.jbusvent.2018.05.006>
- Stewart KJ (2003) Trust transfer on the world wide web. *Organ Sci* 14(1):5–17. <https://doi.org/10.1287/orsc.14.1.5.12810>
- Stiver A, Barroca L, Minocha S, Richards M, Roberts D (2015) Civic crowdfunding research: challenges, opportunities, and future agenda. *New Media Soc* 17(2):249–271. <https://doi.org/10.1177/1461444814558914>
- Talwar S, Talwar M, Kaur P, Dhir A (2020) Consumers' resistance to digital innovations: a systematic review and framework development. *Australas Mark J (AMJ)* 28(4):286–299. <https://doi.org/10.1016/j.ausmj.2020.06.014>
- Tversky A, Kahneman D (1974) Judgment under uncertainty: heuristics and biases: biases in judgments reveal some heuristics of thinking under uncertainty. *Science* 185(4157):1124–1131. <https://doi.org/10.1126/science.185.4157.1124>
- Ulker-Demirel E, Ciftci G (2020) A systematic literature review of the theory of planned behavior in tourism, leisure and hospitality management research. *J Hosp Tour Manag* 43:209–219. <https://doi.org/10.1016/j.jhtm.2020.04.003>
- Vismara S (2016) Equity retention and social network theory in equity crowdfunding. *Small Bus Econ* 46:579–590. <https://doi.org/10.1007/s11187-016-9710-4>
- Vulkan N, Åstebro T, Sierra MF (2016) Equity crowdfunding: a new phenomena. *J Bus Ventur Insights* 5:37–49. <https://doi.org/10.1016/j.jbvi.2016.02.001>
- Wang D, Prokop J (2024) Gender homophily and local bias in equity crowdfunding. *Small Bus Econ*. <https://doi.org/10.1007/s11187-024-00949-5>
- Wolfson JA, Graham DJ, Bleich SN (2017) Attention to physical activity-equivalent calorie information on nutrition facts labels: an eye-tracking investigation. *J Nutr Educ Behav* 49(1):35–42. <https://doi.org/10.1016/j.jneb.2016.10.001>
- Zhao Y, Harris P, Lam W (2019) Crowdfunding industry-history, development, policies, and potential issues. *J Public Aff* 19(1):e1921. <https://doi.org/10.1002/pa.1921>
- Zhu Z, Huang Q, Liu H (2023) How heuristic cues impact crowdfunding performance: the moderating role of platform competition intensity and platform demand potential. *J Bus Res* 160:113797. <https://doi.org/10.1016/j.jbusres.2023.113797>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.